

Q 1

Josephine borrows \$300 000 to buy a unit. Interest is calculated monthly at the rate of 6% per annum compounded monthly. She agrees to repay the loan with equal monthly instalments of M at the end of each month for 20 years. Let A_n be the amount owing after n months.

- Find an expression for A_1 .
- Show that $A_2 = 300\,000(1.005)^2 - M(1.005) - M$.
- Find the amount of the monthly repayment (to nearest 5 cents).

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Q 1
a) $r = 6\% \text{ p.a.}$
 $= 0.06 \text{ p.a.}$
 $= 0.005 \text{ per month}$

i) $A_1 = P(1 + \frac{r}{12}) - M$
 $\therefore A_1 = 300\,000(1.005) - M$

ii) $A_2 = A_1(1 + \frac{r}{12}) - M$

$A_2 = [300\,000(1.005) - M](1.005) - M$
 $= 300\,000(1.005)^2 - M(1.005) - M$
as required.

iii) A_n = amount owing at end of n months

hence $A_{240} = 0$.

$A_{240} = 300\,000(1.005)^{240} - M(1.005)^{239} - M(1.005)^{238} - \dots - M(1.005)^1 - M$

$= 300\,000(1.005)^{240} - M[1 + 1.005 + 1.005^2 + \dots + 1.005^{239}]$
A.P. $a=1, r=1.005$
 $S_n = \frac{a(r^n - 1)}{r - 1}$
 $= \frac{1(1.005^{240} - 1)}{(1.005 - 1)}$

$0 = 300\,000(1.005)^{240} - M \left(\frac{1.005^{240} - 1}{1.005 - 1} \right)$
 $\therefore M = \frac{300\,000(1.005)^{240}(1.005 - 1)}{(1.005^{240} - 1)}$

Q 2

Monique has set up her superannuation fund and after 10 years she has accumulated \$134 000. However due to an accident she is no longer able to work and make further contributions to the fund. Monique is leaving the money in the superannuation fund to accumulate interest at 8% p.a. compounded annually. However she needs to withdraw \$24 000 at the end of each year for normal living expenses.

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E

- Show that at the end of the first year she has \$(134000 \times 1.08 - 24000) in the superannuation fund. 1
- Find a similar expression for the amount in the fund after 3 years. 2
- Hence find how many years the fund will last before there is no money in it. 2

8(b) (i)	$A = P(1+r)^n$ $= 134000 \times (1+0.08)^1$ $= 134000(1.08)$ After 1 year $A_1 = 134000 \times 1.08 - 24000$	1 Mark: Correct answer.
8(b) (ii)	After 2 years $A_2 = (134000(1.08) - 24000) \times 1.08 - 24000$ $= 134000 \times 1.08^2 - 24000(1.08 + 1)$ After 3 years $A_3 = [(134000 \times 1.08^2 - 24000(1.08 + 1)) \times 1.08 - 24000]$ $= 134000 \times 1.08^3 - 24000(1.08^2 + 1.08 + 1)$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution
8(b) (iii)	After n years $A_n = 134000 \times 1.08^n - 24000(1.08^{n-1} + 1.08^{n-2} + \dots + 1)$ To find n when $A_n = 0$ $0 = 134000 \times 1.08^n - 24000 \left(\frac{1.08^n - 1}{1.08 - 1} \right)$ $134000 \times 1.08^n \times 0.08 - 24000(1.08^n - 1) = 0$ $10720 \times 1.08^n - 24000 \times 1.08^n + 24000 = 0$ $13280 \times 1.08^n = 24000$ $1.08^n = \frac{24000}{13280}$ $n \times \log 1.08 = \log \left(\frac{24000}{13280} \right)$ $n \approx 7.69 \text{ years}$	2 Marks: Correct answer. 1 Mark: Obtains a correct expression for A_n and uses $A_n = 0$





Part A: Loan repayments series 2

Q 3

A
C
E

Colin invests \$P at 8% per annum compounded annually. He plans to withdraw \$5000 at the end of each year for six years to cover university fees.

- Write down an expression for the amount A_1 remaining in the account following the withdrawal of the first \$5000.
- Find an expression for the amount A_3 remaining in the account after the third withdrawal.
- How much does Colin need to invest if the account balance is to be \$0 at the end of the six years?

15(c) (i)	$A = P(1+r)^n$ $= P \times (1+0.08)^1$ $= P(1.08)$ After 1 year $A_1 = P(1.08) - 5000$	1 Mark: Correct answer.
15(c) (ii)	After 2 years $A_2 = (P(1.08) - 5000) \times 1.08 - 5000$ $= P \times 1.08^2 - 5000(1.08 + 1)$ After 3 years $A_3 = [(P \times 1.08^2 - 5000(1.08 + 1)) \times 1.08 - 5000]$ $= P \times 1.08^3 - 5000(1.08^2 + 1.08 + 1)$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
15(c) (iii)	After 6 years $A_6 = P \times 1.08^6 - 5000(1.08^5 + 1.08^4 + \dots + 1)$ To find P when $A_6 = 0$ $0 = P \times 1.08^6 - 5000 \left(\frac{1.08^6 - 1}{1.08 - 1} \right)$ $0 = P \times 1.08^6 \times 0.08 - 5000(1.08^6 - 1)$ $P \times 1.08^6 \times 0.08 = 5000(1.08^6 - 1)$ $P = \frac{5000(1.08^6 - 1)}{1.08^6 \times 0.08}$ $= \$23\,114.40$ Colin needs to invest \$23 114.40	2 Marks: Correct answer. 1 Mark: Obtains a correct expression for A_6 and uses $A_6 = 0$.

Q 4

A
R
C

A sum of \$26 000 is borrowed now at 6% per annum reducible interest. Payment is made by n equal annual instalments of \$4 200 beginning at the end of the first year. Find the value of n .

8

$$\begin{aligned}
 A_1 &= 26\,000(1.06) - 4\,200 \\
 A_2 &= A_1(1.06) - 4\,200 \\
 A_2 &= [26\,000(1.06) - 4\,200](1.06) - 4\,200 \\
 &= 26\,000(1.06)^2 - 4\,200(1.06) - 4\,200 \\
 &= 26\,000(1.06)^2 - 4\,200(1.06 + 1) \\
 A_3 &= 26\,000(1.06)^3 - 4\,200(1.06^2 + 1.06 + 1) \\
 A_3 &= 26\,000(1.06)^3 - 4\,200(1 + 1.06 + 1.06^2) \\
 A_n &= 26\,000(1.06)^n - 4\,200(1 + 1.06 + \dots + 1.06^{n-1}) \\
 A_n &= 0 \\
 4\,200(1 + 1.06 + \dots + 1.06^{n-1}) &= 26\,000(1.06)^n \\
 (1 + 1.06 + \dots + 1.06^{n-1}) &\text{ is a GP with } a = 1, r = 1.06, n = n \\
 4\,200 \left(\frac{1(1.06^n - 1)}{1.06 - 1} \right) &= 26\,000(1.06)^n \\
 4\,200 \left(\frac{1.06^n - 1}{0.06} \right) &= 26\,000(1.06)^n \\
 4\,200(1.06)^n - 4\,200 &= 1560(1.06)^n \\
 2\,640(1.06)^n &= 4\,200 \\
 (1.06)^n &= \frac{4200}{2640} \\
 (1.06)^n &= 1.590909 \text{ taking logs} \\
 n \log(1.06) &= \log(1.590909) \\
 n &= \frac{\log 1.590909}{\log 1.06} \\
 n &= 7.9683 \\
 \text{Loan repaid in 8 years}
 \end{aligned}$$

Q 5

ASCHAM

A couple decide to borrow \$350 000 to buy a home. Interest is calculated monthly on the balance still owing, at a rate of 7.2% per annum. The loan is to be repaid at the end of 20 years with equal monthly repayments of \$M. Let the amount left owing after the n th repayment be A_n .

- Derive an expression for A_1 , the amount owing at the end of the first month, just after the first repayment has been made. [1]
- Write an expression involving M for A_{240} , the last repayment [2]
- Show that $M = \frac{2100(1.006)^{240}}{(1.006)^{240} - 1}$ [2]
- Find the value of M to the nearest cent. [1]

$$\begin{aligned} \text{b) i) } A_1 &= 350000(1.006) - M \\ \text{ii) } A_2 &= 350000(1.006)^2 - M(1+1.006) \\ A_{240} &= 350000(1.006)^{240} \\ &\quad - M(1 + 1.006 + \dots + 1.006^{239}) \\ &= 350000(1.006)^{240} - M \frac{(1.006^{240} - 1)}{0.006} \end{aligned}$$

But $A_{240} = 0$

$$\begin{aligned} \text{iii) } M &= \frac{350000(1.006)^{240} \times 0.006}{1.006^{240} - 1} \\ &= \frac{2100(1.006)^{240}}{1.006^{240} - 1} \end{aligned}$$

iv) $M = \$2755.72$

Q 6

ASCHAM

A common strategy for pensioners to access money is to take out a reverse mortgage on their home. A couple is offered \$100,000 from Ditchbank on the following terms:

The loan is to be paid back in forty equal quarterly instalments, the first instalment due at the end of the first quarter. Interest accrues on the balance owing at the rate of 8% per annum, compounded quarterly. Let A_n be the amount owing at the end of the n -th instalment and let \$Q be the size of each instalment.

- Show that the amount owing at the end of the first year is $A_4 = 100000(1.02)^4 - Q(1 + 1.02 + 1.02^2 + 1.02^3)$. 2
- Find the value of Q. 3
- At the end of 5 years, the couple decides to abandon the arrangement. Determine how much they still owe at that stage. 1

a) $n = 40$ $r = 8\% \text{ pa}$

$= 0.02 \text{ p quarter. } \checkmark$

i) $A_1 = 100000(1.02)^1 - Q$ (2)

$A_2 = (100000(1.02)^1 - Q)1.02 - Q$
 $= 100000(1.02)^2 - Q(1.02 + 1) \checkmark$

$A_3 = 100000(1.02)^3 - Q(1.02^2 + 1.02 + 1)$
 $\therefore A_4 = 100000(1.02)^4 - Q(1.02^3 + 1.02^2 + 1.02 + 1)$

ii) $A_{40} = 100000(1.02)^{40} - Q(1.02^{39} + 1.02^{38} + \dots + 1) \checkmark$
but $A_{40} = 0$

$\therefore Q \left(\frac{1(1.02^{40} - 1)}{1.02 - 1} \right) = 100000(1.02)^{40} \checkmark$

$\therefore Q = \frac{100000(1.02)^{40}}{\frac{1((1.02)^{40} - 1)}{0.02}}$ (3)

$\hat{=} 3656$

iii) 5 years $A_{20} = 100000(1.02)^{20} - 3655.57 \left(\frac{1(1.02^{20} - 1)}{1.02 - 1} \right) \checkmark$
Owing $\hat{=} \$59774$. (1)



Part A: Loan repayments series 2



Q 7

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A
M

Diana borrows \$10 000 and arranges to pay it back with interest in 20 equal instalments every three months over 5 years. She is charged 6% per annum interest compounded monthly.

Let A_n be the amount owing after n months and let M be the instalment.

- Find the amount owing after the first three months, just after she has made her first payment. (1)
- Show that $M = \frac{10000(1.005^{63} - 1.005^{60})}{1.005^{60} - 1}$. (3)
- Find the size of the instalment she pays. (1)
- How much does Diana pay in interest? (1)

a) \$10 000

Every 3 months 5 years

$\therefore$ 20 payments

6% pa = 0.005 p/month

$$i) A_1 = 10000(1.005) \\ A_3 = 10000(1.005)^3 - M \quad (1)$$

$$ii) A_6 = A_3(1.005)^3 - M \\ = [10000(1.005)^3 - M]1.005^3 - M \\ = 10000(1.005^6) - M(1 + 1.005^3)$$

$$\therefore A_{60} = 10000(1.005)^{60} - M(1 + 1.005^3 + \dots + 1.005^{57}) \\ A_{60} = 10000(1.005)^{60} - M \frac{(1.005^{60} - 1)}{1.005^3 - 1}$$

$$A_{60} = 0 \\ 10000(1.005)^{60} = \frac{M(1.005^{60} - 1)}{1.005^3 - 1}$$

$$\frac{10000(1.005)^{60}(1.005^3 - 1)}{1.005^{60} - 1} = M$$

$$M = \frac{10000(1.005^{63} - 1.005^{60})}{1.005^{60} - 1} \quad (3)$$

$$iii) M = \$582.89 \quad (1)$$

$$v) \text{Interest} = 582.89 \times 20 - 10000 \\ = \$1657.80 \quad (1)$$

Q 8

B
A
R
K
E
R

Rachael borrows \$P for an overseas holiday.

This loan plus interest is to be repaid in equal monthly instalments of \$232 over five years. Interest of 6% p.a. is charged monthly on the balance owing at the start of each month.

Let A_n be the amount owing at the end of n months.

- Show that $A_2 = P(1.005)^2 - 232(1 + 1.005)$ 2
- Prove that $A_n = P(1.005)^n - 46400(1.005^n - 1)$ 2
- Hence, or otherwise, find the amount that Rachael borrowed. 2

$$c) A_1 = P(1.005) - 232$$

$$i) A_2 = [P(1.005) - 232](1.005) - 232 \\ = P(1.005)^2 - 232(1.005) - 232 \\ = P(1.005)^2 - 232(1 + 1.005)$$

$$ii) A_n = P(1.005)^n - 232(1 + 1.005 + \dots + 1.005^{n-1}) \\ = P(1.005)^n - \frac{232 \times 1(1.005^n - 1)}{1.005 - 1} \\ = P(1.005)^n - 46400(1.005^n - 1)$$

$$iii) A_{60} = 0 \text{ amt paid back} \\ \therefore P(1.005)^{60} - 46400(1.005^{60} - 1) = 0 \\ P = \frac{46400(1.005^{60} - 1)}{1.005^{60}}$$

$$= \$12000.33 \\ \approx \$12000$$



Q 9

B
A
R
K
E
R

Amelia has borrowed \$5000 at the beginning of 2003.
The debt is to be repaid by equal annual installments of \$1200.
The first installment is to be repaid at the beginning of 2004.
Interest at the rate of 18% p.a. is calculated at the beginning of each year on the balance owing at the end of the previous year.
This interest is then added to the balance of the debt before a repayment is made.

- Calculate the amount owing on the debt after the first repayment is made at the beginning of 2004. 2
- Amelia wants to clear her debt at the beginning of 2010. 3
How much extra will she have to repay at the beginning of 2010 (after making her normal repayment) in order for her to do this?

$$(i) (c) (i) A_1 = 5000 \times 1.18 - 1200$$

$$= \$4700$$

$$(ii) A_2 = A_1 \times 1.18 - 1200$$

$$= 5000 \times 1.18^2 - 1200(1.18 + 1)$$

$$A_3 = A_2 \times 1.18 - 1200$$

$$= 5000 \times 1.18^3 - 1200(1.18^2 + 1.18 + 1)$$

$$\text{begin of } 2010 \Rightarrow A_7 = 5000 \times 1.18^7 - 1200(1.18^6 + \dots + 1.18 + 1)$$

$$A_7 = 5000 \times 1.18^7 - 1200 \times \frac{1.18^7 - 1}{1.18 - 1} = \$1357.54$$

G.P. with $a=1, r=1.18, n=7$

Q10

B
A
R
K
E
R

Robert takes out a \$250 000 mortgage and agrees to pay the bank a \$2000 installment each month. The interest rate on the loan is 7.2% per annum, compounded monthly, and the contract requires that the loan is paid off within 20 years.

- Find the amount owing on the loan at the end of the 10th year. 1
- Find A_{240} and explain why this shows that the loan is actually paid off in less than 20 years. 2
- Find how many months early that the loan is paid off. 4
(ie. how many months below 20 years)

$$(i) A_{120} = 250000(1.006)^{120} - \frac{2000(1.006^{120} - 1)}{0.006}$$

$$= 162\,498.4953$$

$$= \$162\,498.50 \text{ (2dp)}$$

$$(ii) A_{240} = 250000(1.006)^{240} - \frac{2000(1.006^{240} - 1)}{0.006}$$

$$= -16\,881.16943$$

Since A_{240} is negative, then Robert has paid more than he owed,
ie. loan is in credit after 20 yrs.
The loan was paid off in less than 20 years.

$$(iii) A_n = 0$$

$$0 = 250000(1.006)^n - \frac{2000(1.006^n - 1)}{0.006}$$

$$= 250000(1.006)^n - 333333.3(1.006)^n + 333333.3$$

$$= 333333.3 - 83333.3(1.006)^n$$

$$83333.3(1.006)^n = 333333.3$$

$$(1.006)^n = \frac{333333.3}{83333.3}$$

$$1.006^n = 4$$

$$n \ln 1.006 = \ln 4$$

$$n = \frac{\ln 4}{\ln 1.006}$$

$$= 231.7415 \dots$$

$$= 232 \text{ months}$$

$\therefore 240 - 232 = 8 \text{ months}$
Loan was paid off 8 months early.

Q11

BARKER

- (i) Greg borrows \$400 000 in order to buy an apartment. The interest rate is 6% p.a. reducible and the loan is to be repaid in equal monthly repayments of \$M over 25 years with the interest calculated monthly. Let \$A_n be the amount owing after the nth repayment.

- (α) Write down expressions for \$A₁ and \$A₂, the amounts owing after the first and second repayments have been made respectively.
- (β) Show that the amount of each monthly repayment is \$2577.21 (correct to the nearest cent).

- (ii) After 5 years (i.e. 60 repayments) the interest rate rises to 9% p.a. Find the new monthly repayment \$N, correct to the nearest cent. (Assume that the period of the loan is still 25 years).
- (iii) How much extra does Greg repay over the life of the loan as a result of the 3% p.a. interest rate rise?

(6)(i) (α)

$$A_1 = 400\,000 \times 1.005 - M$$

$$A_2 = 400\,000 \times 1.005^2 - 1.005M - M$$

$$(β) A_{300} = 400\,000 \times 1.005^{300} - M(1 + 1.005 + \dots + 1.005^{299})$$

But $A_{300} = 0$, so

$$M = \frac{400\,000 \times 1.005^{300}}{(1.005^{300} - 1)}$$

$$= 2577.205606$$

$$= \$2577.21 \text{ (n. cent)}$$

$$(ii) A_{60} = 400\,000 \times 1.005^{60} - 2577.21(1 + 1.005 + \dots + 1.005^{59})$$

$$= 400\,000 \times 1.005^{60} - 2577.21 \times \frac{1.005^{60} - 1}{0.005}$$

$$= 359\,728.0407$$

$$= \$359\,728.04 \text{ (n. cent)}$$

$$A_{61} = 359\,728.04 \times 1.0075 - N$$

$$A_{62} = 359\,728.04 \times 1.0075^2 - 1.0075N - N$$

$$\vdots$$

$$A_{300} = 359\,728.04 \times 1.0075^{240} - N(1 + 1.0075 + \dots + 1.0075^{239})$$

But $A_{300} = 0$, so

$$N = \frac{359\,728.04 \times 1.0075^{240}}{(1.0075^{240} - 1)}$$

$$= 3236.566546$$

$$\therefore \text{Req'd amt} = \$3236.57 \text{ (n. cent)}$$

$$(iii) \text{ Extra paid} = \$ (3236.57 - 2577.21) \times 240$$

$$= \$158\,246.40$$

Q12

BAULKHAM

Kelvin borrows \$200 000 from his bank. Interest is compounded monthly at 0.425% per month. A_n is the amount owed after n payments, \$M is the amount of the monthly instalments and the loan is repaid after n months.

(i) Show that $A_2 = 200\,000(1.00425)^2 - M(1.00425) - M$

(ii) Show that the monthly repayment, \$M is given by

$$M = \frac{200\,000(1.00425)^n(0.00425)}{1.00425^n - 1}$$

(iii) Find the amount of the monthly instalments if Kelvin agrees to repay the loan over 30 years.

(iv) How much will Kelvin pay in total after 30 years?

(v) If Kelvin instead decided to pay monthly instalment of \$1331 from the beginning of the loan, how long will he take to repay the loan?

3(i) $P = \$200\,000$
 $r = 0.425\%$

$$A_1 = 200\,000(1.00425) - M$$

$$A_2 = A_1(1.00425) - M$$

$$= [200\,000(1.00425) - M] \times 1.00425 - M$$

$$= 200\,000(1.00425)^2 - M(1.00425) - M$$

$$A_3 = A_2(1.00425) - M$$

$$= [200\,000(1.00425)^2 - M(1.00425) - M] \times 1.00425 - M$$

$$= 200\,000(1.00425)^3 - M(1.00425)^2 - M(1.00425) - M$$

$$A_n = 200\,000(1.00425)^n - M(1.00425)^{n-1} - \dots - M(1.00425) - M$$

$$A_n = 200\,000(1.00425)^n - M \left(\frac{1.00425^n - 1}{0.00425} \right)$$

u.p with a=1
n=1.00425

If the loan is paid after n months

$$A_n = 0$$

$$M \left(\frac{1.00425^n - 1}{0.00425} \right) = 200\,000(1.00425)^n$$

$$M = \frac{200\,000(1.00425)^n(0.00425)}{1.00425^n - 1}$$

3(ii) 30 years = 360 months
= 360 instalments

$$M = \frac{200\,000(1.00425)^{360}(0.00425)}{1.00425^{360} - 1}$$

$$= \$1085.90$$

4(i) total repaid = 360×1085.90
 $= 390924$

5(i) $1331 = \frac{200\,000(1.00425)^h \times 0.00425}{1.00425^h - 1}$

$$1331(1.00425)^h - 1331 = 850(1.00425)^h$$

$$(1331 - 850)(1.00425)^h = 1331$$

$$(1.00425)^h = 2.767151 \dots$$

$$h \ln(1.00425) = \ln(2.767151 \dots)$$

$$h = \frac{\ln(2.767151 \dots)}{\ln(1.00425)}$$

$$= 239.99 \dots = 240 \text{ months}$$

(2 marks with correct ans & working)



Part A: Loan repayments series 2

Q11

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A
M

Sonia and Jenny want to save a deposit of \$50 000 to buy a house. They devise a savings plan to allow them to achieve this goal. Beginning on 1st January 2014, they deposit \$1000 on the first day of each month into an account which pays 9% p.a compounded monthly.

- Find the amount they have in the account on 31st January 2014. 1
(i.e. at the end of the first month)
- Show that the amount they have in the account at the end of n months is given by $A = \frac{1007.5(1.0075^n - 1)}{0.0075}$ 2
- Hence find the least number of months they need to save their deposit. 2

$$\begin{aligned} \text{b) i) } A_1 &= 1000(1 + \frac{0.09}{12})^1 \\ &= 1000(1.0075) \\ &= \$1007.50 \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii) } A_2 &= 1000(1.0075)^2 \\ A_3 &= 1000(1.0075)^3 \\ &\vdots \\ A_n &= 1000(1.0075)^n \\ \therefore A &= 1000(1.0075) + 1000(1.0075)^2 + \dots + (1.0075)^n \\ &= 1000[1.0075 + 1.0075^2 + \dots + 1.0075^n] \checkmark \end{aligned}$$

$$\begin{aligned} &= 1000 \left[\frac{1.0075(1.0075^n - 1)}{1.0075 - 1} \right] \checkmark \\ &= 1007.50 \left(\frac{1.0075^n - 1}{0.0075} \right) \end{aligned}$$

$$\text{(iii) } 50000 = 1007.5 \left(\frac{1.0075^n - 1}{0.0075} \right)$$

$$\begin{aligned} 0.3722 &= 1.0075^n - 1 \\ 1.3722 &= 1.0075^n \\ \log 1.3722 &= n \log 1.0075 \\ n &= \frac{\log 1.3722}{\log 1.0075} \\ n &= 42.347 \\ \therefore \text{least no. of months} &= 43 \checkmark \end{aligned}$$

Q12

C
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B
A
H

Ella borrowed \$180 000 to finance an extension on her home. She agreed to pay off the loan in equal monthly instalments of \$P, paid at the end of each month, at an interest rate of 6% per annum, compounded monthly.

- Show that after the first instalment is paid, the amount owing on the loan is: 1
 $\$[180\,000(1.005) - P]$
- Show that after three months she owes: 2
 $\$[180\,000(1.005)^3 - P((1.005)^2 + (1.005) + 1)]$
- If the loan is repaid after 8 years, find the value of P, the monthly instalment. 3

$$\text{9) a) } \$180\,000 \quad \$P/\text{month} \quad 6\% \text{ pa} = 0.5\% \text{ per month}$$

$$\begin{aligned} \text{i) } A_1 &= 180\,000 + 180\,000 \times \frac{0.5}{100} - P \\ &= 180\,000(1 + 0.005) - P \\ &= 180\,000(1.005) - P \end{aligned}$$

$$\begin{aligned} \text{ii) } A_2 &= A_1 \times 1.005 - P \\ &= [180\,000 \times 1.005 - P] \times 1.005 - P \\ &= 180\,000 \times 1.005^2 - P \times 1.005 - P \\ &= 180\,000 \times 1.005^2 - P(1.005 + 1) \end{aligned}$$

$$\begin{aligned} A_3 &= A_2 \times 1.005 - P \\ &= [180\,000 \times 1.005^2 - P(1.005 + 1)] \times 1.005 - P \\ &= 180\,000 \times 1.005^3 - P(1.005^2 + 1.005) - P \\ &= 180\,000 \times 1.005^3 - P(1 + 1.005 + 1.005^2) \end{aligned}$$

$$\text{iii) } n = 8 \text{ years} = 96 \text{ months}$$

$$\therefore A_{96} = 0$$

$$180\,000 \times 1.005^{96} - P(1 + 1.005 + 1.005^2 + \dots + 1.005^{95}) = 0$$

$$P = \frac{180\,000 \times 1.005^{96}}{1 + 1.005 + \dots + 1.005^{95}}$$

$$\uparrow \text{AP } a=1, r=1.005, n=96$$

$$P = \frac{180\,000 \times 1.005^{96}}{\frac{1(1.005^{96} - 1)}{1.005 - 1}} = \frac{180\,000 \times 1.005^{96} \times 0.005}{1.005^{96} - 1}$$

$$P = \$2365.46$$

Q13

B
A
U
L
K
H
A
M

A dealership offers Ashwin \$20000 loan for a new car at the beginning of the year for which it charges interest at the rate of 6%pa.

- (i) Ashwin wants to make monthly repayments at the end of the month and pay it off in 5 years.

Let M be Ashwin's repayments

- (a) Show that the balance of Ashwin's account at the end of the 2nd month is
 $A_2 = 20000 \times 1.005^2 - 1.005M - M$

- (b) Find an expression for the amount owed at the end of the 5 years and deduce that

$$M = \frac{0.005(20000 \times 1.005^{60})}{1.005^{60} - 1} = \$387 \text{ (to the nearest dollar)}$$

- (c) Show that at the end of the first year Ashwin will owe the dealership \$16460 (to the nearest ten dollars)

- (ii) At the beginning of the 2nd year, Ashwin's mother decides to secretly make \$101 repayments at the beginning of the month to help him pay it off earlier.

How many payments did Ashwin's mother have to make until it was paid off?

d) $P = 20000$ $R = 0.06 \text{ pa}$ $n = 5 \text{ yrs}$
 $R = 0.005 \text{ per month}$ $n = 60 \text{ months}$

(a) $A_1 = 20000 \times 1.005 - M$

$A_2 = 20000 \times 1.005^2 - 1.005M - M$

(b) $A_60 = 20000 \times 1.005^{60} - 1.005^{59}M - \dots - M$

$0 = 20000 \times 1.005^{60} - M(1.005^{59} + \dots + 1)$

$20000 \times 1.005^{60} = M \left[\frac{1(1.005^{60} - 1)}{1.005 - 1} \right]$

$0.005(20000 \times 1.005^{60}) = M(1.005^{60} - 1)$

$\therefore M = \frac{0.005(20000 \times 1.005^{60})}{1.005^{60} - 1}$

$= \$387$

(c) $A_{12} = 20000 \times 1.005^{12} - 387 \left(\frac{1.005^{12} - 1}{1.005 - 1} \right)$
 $A_{12} = \$16459.693 \dots$
 $A_{12} \approx \$16460$

ii) $A_1 = 16460 \times 1.005 - 101 \times 1.005 - 387$ or $A_1 = (16460 - 101) \times 1.005 - 387$

$A_2 = 16460 \times 1.005^2 - 101 \times 1.005^2 - 1.01 \times 1.005 - 387 \times 1.005 - 387$

$A_n = 16460 \times 1.005^n - 101 \times 1.005^n - 101 \times 1.005^{n-1} - \dots - 101 \times 1.005 - 387 \times 1.005^{n-1} - \dots - 387$

$0 = 16460 \times 1.005^n - 101(1.005^n + 1.005^{n-1} + \dots + 1.005) - 387(1.005^{n-1} + 1.005^{n-2} + \dots + 1)$

$0 = 16460 \times 1.005^n - 101 \left[\frac{1.005(1.005^n - 1)}{1.005 - 1} \right] - 387 \left[\frac{1(1.005^n - 1)}{1.005 - 1} \right]$

$0 = 16460 \times 1.005^n - 20301(1.005^n - 1) - 77400(1.005^n - 1)$

$0 = 16460 \times 1.005^n - 97701(1.005^n - 1)$

$0 = 16460 \times 1.005^n - 97701 \times 1.005^n + 97701$
 $- 97701 = -81241 \times 1.005^n$

$97701 = 1.005^n \times 81241$

$n = \log_{1.005} \left(\frac{97701}{81241} \right)$

$n = 36.99 \dots$

$\therefore 37 \text{ payments}$



Part A: Loan repayments series 2



Q14

C
A
R
I
N
G
B
A
H

At the completion of his degree, Oliver had a HEC's debt of \$100 000. He plans to repay this in equal monthly repayments of \$M. Interest is charged at a rate of 0.5% per month.

- (i) Show that the amount owing after 3 months, A_3 , is given by

$$A_3 = 100\,000 \times 1.005^3 - M(1 + 1.005 + 1.005^2)$$

- (ii) If Oliver wishes to pay off his loan by the end of 10 years, then show that he will need to pay \$1 110 per month.

- (iii) Show that the amount owing after n months can be written as,

$$A_n = 1.005^n [100\,000 - 200M] + 200M$$

- (iv) If Oliver decides that he can only repay \$750 each month, how long will it take him to repay the loan? (Answer in years and months)

$$\begin{aligned} i) A_1 &= 100\,000 \times 1.005 - M \\ A_2 &= 100\,000 \times 1.005^2 - M \times 1.005 - M \\ A_3 &= 100\,000 \times 1.005^3 - M \times 1.005^2 - M \times 1.005 - M \\ &= 100\,000 \times 1.005^3 - M(1 + 1.005 + 1.005^2) \end{aligned}$$

$$ii) 10 \text{ yrs} \Rightarrow n = 120$$

$$\begin{aligned} A_n &= 100\,000 \times 1.005^n - M(1 + 1.005 + \dots + 1.005^{n-1}) \\ M &= \frac{100\,000 \times 1.005^{120} - A_{120}}{\frac{1.005^{120} - 1}{0.005}} \\ &= \$1\,110.20 \end{aligned}$$

$$\begin{aligned} iii) A_n &= 100\,000 \times 1.005^n - M \left[\frac{1.005^n - 1}{0.005} \right] \\ &= 100\,000 \times 1.005^n - 200M [1.005^n - 1] \\ &= 1.005^n [100\,000 - 200M] + 200M \end{aligned}$$

$$\begin{aligned} iv) 1.005^n [100\,000 - 200 \times \$750] + 200 \times \$750 &= 0 \\ 1.005^n &= \frac{-200 \times 750}{100\,000 - 200 \times 750} \\ &= \frac{150\,000}{50\,000} \end{aligned}$$

$$\begin{aligned} n &= \frac{\ln 3}{\ln 1.005} \\ &= 220 \text{ mths} \\ &= 18 \text{ yrs } 5 \text{ mths} \end{aligned}$$

Q15

C
A
R
I
N
G
B
A
H

Paula borrows \$15 000 at 4% per quarter reducible interest. She pays the loan off over 5 years by paying quarterly repayments of \$R.

Let A_n be the amount of money Paula still owes after the n th repayment is made.

- (i) Write an expression for A_1 .

- (ii) Show that $A_n = 15\,000 \times 1.04^n - R(1.04^{n-1} + \dots + 1.04 + 1)$

- (iii) Hence find the value of R .

$$i) A_1 = 15\,000(1.04) - R$$

$$\begin{aligned} ii) A_2 &= (15\,000(1.04) - R)1.04 - R \\ &= 15\,000(1.04)^2 - 1.04R - R \end{aligned}$$

$$\begin{aligned} A_n &= 15\,000(1.04)^n - 1.04^{n-1}R - \dots - R \\ &= 15\,000(1.04)^n - R[1.04^{n-1} + \dots + 1] \end{aligned}$$

- (iii) After 20 repayments \$0 is owed.

$$0 = 15\,000(1.04)^{20} - R(1.04^{19} + \dots + 1)$$

$$\begin{aligned} a &= 1 \\ r &= 1.04 \\ n &= 20 \\ S_{20} &= 1 \left(\frac{1.04^{20} - 1}{1.04 - 1} \right) \\ &= 29.778 \dots \end{aligned}$$

$$\begin{aligned} R &= \frac{15\,000(1.04)^{20}}{29.778 \dots} \\ &= \$1\,103.73 \end{aligned}$$



Part A: Loan repayments series 2



Q16

C
R
A
N
B
R
O
O
K

John and Helen are two farmers. They borrow \$400 000 from a bank. They make monthly repayments and the interest is 6% p.a. compounded monthly. The loan is for 20 years. Because of a drought the bank allow them to begin repaying the loan at the end of the fourth month. Let A_n be the amount owing at the end of n months and m the monthly repayments.

(i) Show that $A_5 = 400\,000(1.005)^5 - 2.005m$.

2

(ii) Find the monthly repayment m .

3

(i) Amount owing after 1 month, $A_1 = P(1.005)$
where $P = 400\,000$

Similarly, amount owing after 2 months,
 $A_2 = P(1.005)^2$
and $A_3 = P(1.005)^3$ ✓
and $A_4 = P(1.005)^4 - m$
and $A_5 = (P(1.005)^4 - m)1.005 - m$
 $= P(1.005)^5 - m(1 + 1.005)$
 $= P(1.005)^5 - 2.005m$ ✓

Now $A_6 = A_5(1.005) - m$
 $= [P(1.005)^5 - m(1 + 1.005)]1.005 - m$
 $\therefore A_6 = P(1.005)^6 - m(1 + 1.005 + 1.005^2)$

Now continuing this pattern $\therefore$ amount owing after 20 years is 240 months

$A_{240} = P(1.005)^{240} - m(1 + 1.005 + 1.005^2 + \dots + 1.005^{236})$
But as loan is repaid after 20 years ✓
 $A_{240} = 0$
 $\therefore m = \frac{P(1.005)^{240}}{1 + 1.005 + \dots + 1.005^{236}}$
A.P. $a=1, r=1.005, n=237$

$\therefore m = \frac{400\,000(1.005)^{240}}{1 \left[\frac{1.005^{237} - 1}{0.005} \right]}$ ✓
 $= 2928.03$ (2dp)

$\therefore$ monthly repayment, $m = \$2928.03$ (to nearest cent)

Q17

C
R
A
N
B
R
O
O
K

Hanna wishes to buy a used car that is on sale for \$12 000. The car salesman offers her the following deal. The first 3 months she will not have to pay any interest but the first repayment of \$ M will be due at the end of the first month. Interest will be charged at 1% a month, calculated before the monthly repayment is made. If Hanna plans to pay back the loan in 6 years;

i. Find an expression for A_3 .

1

ii. Show that $A_5 = (12000 - 3M)(1.01)^2 - M(1 + 1.01)$

2

iii. Hence write down an expression for A_{72}

1

iv. Find the value of M

2

(i) $A_0 = 12000$

$A_1 = 12000 - M$

$A_3 = 12000 - 3M$

(ii) $A_4 = (12000 - 3M) \times 1.01 - M$

$A_5 = [(12000 - 3M) \times 1.01 - M] \times 1.01 - M$

$= (12000 - 3M) \times 1.01^2 - M \times (1.01 + 1)$

(iii) $A_{72} = (12000 - 3M) \times 1.01^{69} - M(1.01^{68} + 1.01^{67} + \dots + 1.01 + 1)$

(iv) Since $A_{72} = 0 = (12000 - 3M) \times 1.01^{69} - M \left(\frac{1.01^{69} - 1}{1.01 - 1} \right)$

$M \left(\frac{1.01^{69} - 1}{0.01} + 3 \times 1.01^{69} \right) = 12000 \times 1.01^{69}$

$M = \frac{12000 \times 1.01^{69}}{\frac{1.01^{69} - 1}{0.01} + 3 \times 1.01^{69}}$

$= \$227.83$



Q18

C
R
A
N
B
R
O
O
K

Geoff borrows \$350 000 from a bank for 20 years at 6% p.a. compounded monthly. The conditions of the loan are that there will be no interest charged for the first three months but repayments will be required at the end of every month. The first interest charged is at the end of the fourth month.

Let A_n be the amount owing at the end of n months and let m be the monthly repayments.

- Build an expression for A_4 . 2
- Show that $A_5 = (350\,000 - 3m)1.005^2 - m(1 + 1.005)$. 2
- Find the monthly repayments if the loan is to be repaid in 20 years. 4

$$(i) A_1 = 350\,000 - m \quad (r = 0.005)$$

$$A_2 = 350\,000 - 2m$$

$$A_3 = 350\,000 - 3m$$

$$A_4 = (350\,000 - 3m)(1.005) - m$$

$$(ii) A_5 = A_4(1.005) - m$$

$$\therefore A_5 = (350\,000 - 3m)(1.005)^2 - m(1.005) - m$$

$$\therefore A_5 = (350\,000 - 3m)(1.005)^2 - m(1 + 1.005)$$

$$(iii) 20 \text{ years} = 240 \text{ months.}$$

$$A_5 = (350\,000 - 3m)(1.005)^4 - m(1 + 1.005)$$

$$A_{240} = (350\,000 - 3m)(1.005)^{237} - m(1 + 1.005 + \dots + 1.005^{236})$$

$$A_{240} = (350\,000 - 3m)(1.005)^{237} - m \left[\frac{1(1.005^{237} - 1)}{1.005 - 1} \right]$$

$$A_{240} = (350\,000 - 3m)(1.005)^{237} - m \left[\frac{1.005^{237} - 1}{0.005} \right]$$

$$0 = (350\,000)(1.005)^{237} - (3m)(1.005)^{237} - m \left[\frac{1.005^{237} - 1}{0.005} \right]$$

$$3m(1.005)^{237} + m \left[\frac{1.005^{237} - 1}{0.005} \right] = 350\,000(1.005)^{237}$$

$$m \left[3(1.005)^{237} + \left(\frac{1.005^{237} - 1}{0.005} \right) \right] = 350\,000(1.005)^{237}$$

$$\therefore m = \frac{350\,000(1.005)^{237}}{3(1.005)^{237} + \left(\frac{1.005^{237} - 1}{0.005} \right)}$$

$$\therefore m = \$2470.53$$

Q19

C
S
S
A

Mr. Egan borrows \$P from a bank to fund his house extensions. The term of the loan is 20 years with an annual interest rate of 9%. Each month, interest is calculated on the balance at the beginning of the month and added to the balance owing. Mr. Egan repays the loan in equal monthly instalments of \$1 050.

- Write an expression for the amount, A_1 , Mr. Egan owes immediately at the end of the first month. 1
- Show that at the end of n months the amount owing, A_n , is given by: 3

$$A_n = P(1.0075)^n - 140\,000(1.0075)^n + 140\,000$$
- If at the end of 20 years the loan has been repaid, calculate the amount Mr. Egan originally borrowed, correct to the nearest dollar. 2

(c) (i) (1 mark)

$$A_1 = P \left(1 + \frac{0.09}{12} \right) - 1050 \therefore A_1 = P(1.0075) - 1050$$

(c) (ii) (3 marks)

$$A_2 = [P(1.0075) - 1050](1.0075) - 1050$$

$$= P(1.0075)^2 - 1050(1.0075) - 1050$$

$$= P(1.0075)^2 - 1050(1 + 1.0075)$$

$$A_3 = P(1.0075)^3 - 1050[1 + 1.0075 + 1.0075^2]$$

$$A_n = P(1.0075)^n - 1050(1 + 1.0075 + \dots + 1.0075^{n-1})$$

$$= P(1.0075)^n - 1050 \left[\frac{1(1.0075^n - 1)}{0.0075} \right]$$

$$= P(1.0075)^n - 140\,000 [1.0075^n - 1]$$

$$= P(1.0075)^n - 140\,000(1.0075)^n + 140\,000$$

(c) (iii) (2 marks)

When $n = 240$

$$\therefore 0 = P(1.0075)^{240} - 140\,000(1.0075)^{240} + 140\,000$$

$$\therefore P(1.0075)^{240} = 140\,000(1.0075)^{240} - 140\,000$$

$$\therefore P = \frac{140\,000(1.0075)^{240} - 140\,000}{(1.0075)^{240}}$$

$$\therefore P = \$116\,702 \text{ (to the nearest dollar)}$$



Q20

C
S
S
A

Another sister city of Sydney is San Francisco. Sydney City Council decides to build an art gallery in San Francisco to allow local Sydney artists to exhibit their work.

The loan required to build the art gallery is \$ P with interest charged at an introductory rate of 6% p.a. for the first three months. The loan is to be initially repaid in equal monthly repayments of \$4 000 over 3 years and interest is charged monthly before each repayment.

Let S_n be the amount owing by Sydney City Council at the end of the n th repayment

- Find an expression for A_1 . 1
 - Show that $A_3 = P(1.005)^3 - 4000(1 + 1.005 + 1.005^2)$. 2
- At the end of three months interest rates rise to 9%p.a. and the loan is to be repaid in total in equal monthly repayments of \$4 800 for the next 2.75 years.
- If the loan's interest rate is fixed at 9%p.a. for the remainder of the loan, find the value of P . 2

(b) (i) (1 mark)

$$A_1 = P(1.005) - 4000$$

(b) (ii) (2 marks)

$$A_1 = P \times (1.005) - 4000$$

$$A_2 = (A_1 \times 1.005) - 4000 = [(P \times (1.005) - 4000) \times 1.005 - 4000]$$

$$= P(1.005)^2 - 4000(1 + 1.005)$$

$$A_3 = (A_2 \times 1.005) - 4000 = [P(1.005)^2 - 4000(1 + 1.005)] \times 1.005 - 4000$$

$$= P(1.005)^3 - 4000(1 + 1.005 + 1.005^2)$$

(b) (iii) (2 marks)

Continuing the pattern,

$$A_4 = A_3 \times 1.0075 - 4800$$

$$A_5 = [A_4(1.0075) - 4800] \times 1.0075 - 4800 = A_3(1.0075)^2 - 4800(1 + 1.0075)$$

$$A_6 = [A_5(1.0075) - 4800] \times 1.0075 - 4800 = A_3(1.0075)^3 - 4800(1 + 1.0075 + 1.0075^2)$$

...

$$A_{36} = A_3(1.0075)^{33} - 4800(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{32})$$

but $A_{36} = 0$ as the loan is repaid in full

$$\therefore 0 = A_3(1.0075)^{33} - 4800 \left[\frac{1(1.0075^{33} - 1)}{1.0075 - 1} \right]$$

$$\therefore A_3 = \frac{640000(1.0075^{33} - 1)}{1.0075^{33}} \text{ but } A_3 = P(1.005)^3 - 12060.10$$

$$\therefore P(1.005)^3 = \frac{640000(1.0075^{33} - 1)}{1.0075^{33}} + 12060.10$$

$$P = \$149\,662.11$$

Q21

C
S
S
A

Reymark borrows \$50 000 at the beginning of 2013 from his local Building Society. The loan is to be repaid in equal monthly repayments of \$900, with interest charged at 7.2% p.a. at the end of each month, just before each repayment.

Let A_n be the amount owing after the n th repayment.

- Show that $A_n = 50\,000(1.006)^n - 900(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$. 2
- After how many months will Reymark have halved his loan? 3

(c) (i) (2 marks)

$$A_1 = 50000 \left(1 + \frac{0.6}{100} \right) - 900 = 50000(1.006) - 900$$

$$A_2 = 50000(1.006)^2 - 900(1 + 1.006)$$

$$A_3 = 50000(1.006)^3 - 900(1 + 1.006 + 1.006^2) \dots$$

$$\therefore A_n = 50000(1.006)^n - 900(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$

(c) (ii) (3 marks)

Let $A_n = 25000$

$$\therefore 25000 = 50000(1.006)^n - 900(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$

$$\therefore 25000 = 50000(1.006)^n - 900 \left(\frac{1.006^n - 1}{0.006} \right)$$

$$\therefore 25000 = 50000(1.006)^n - 150000(1.006^n - 1) \quad (+25000)$$

$$\therefore 1 = 2(1.006)^n - 6(1.006^n - 1) \Rightarrow 1 = 2(1.006)^n - 6(1.006)^n + 6$$

$$\therefore 4(1.006)^n = 5 \Rightarrow n = \frac{\ln \frac{5}{4}}{\ln 1.006} = 37.3$$

After 37.3 months Reymark will have halved his loan.

Q22

C
S
S
A

Helen borrows \$30 000 over 4 years to purchase a 4WD from a car dealership.

The dealer offers an "interest free" period for the first 6 months of the loan.

After 6 months, the remainder of the loan is charged at 18%p.a. with interest calculated each month, just before each repayment.

The loan is to be repaid in 48 equal monthly repayments of \$M.

Let A_n be the amount owing after the n th repayment.

- Find an expression for A_6
- Show that $A_8 = (30\ 000 - 6M)(1.015)^2 - M(1 + 1.015)$.
- Find the value of Helen's monthly repayment \$M.

(b) (i) (1 mark)

Sample answer:

$$A_1 = 30\ 000 - M$$

$$A_2 = 30\ 000 - 2M$$

$$A_3 = 30\ 000 - 3M$$

$$\therefore A_6 = 30\ 000 - 6M$$

(b) (ii) (2 marks)

Sample answer:

$$A_7 = A_6(1.015) - M = [30000 - 6M](1.015) - M$$

$$A_8 = A_7(1.015) - M = [(30000 - 6M)(1.015) - M](1.015) - M$$

$$A_8 = (30000 - 6M)(1.015)^2 - M(1.015) - M$$

$$\therefore A_8 = (30000 - 6M)(1.015)^2 - M(1 + 1.015)$$

(b) (iii) (2 marks)

Sample answer:

When $n = 48$

$$\therefore A_{48} = (30000 - 6M)(1.015)^{42} - M(1 + 1.015 + 1.015^2 + \dots + 1.015^{41})$$

$$0 = (30000 - 6M)(1.015)^{42} - M \left(\frac{1.015^{42} - 1}{1.015 - 1} \right) \Rightarrow M \left(\frac{1.015^{42} - 1}{1.015 - 1} \right) = 30000(1.015)^{42} - 6M(1.015)^{42}$$

$$M \left[\frac{1.015^{42} - 1}{0.015} + 6(1.015)^{42} \right] = 30000(1.015)^{42}$$

$$M[57.923141 + 11.21308269] = 56065.41345$$

$$\therefore M \approx \$811$$

Q23

F
O
R
T
S
T

A loan of \$1000 is to be repaid by equal annual instalments, repayments commencing at the end of the first year of the loan. Interest, at the rate of 10 per cent, is calculated each year on the balance before each repayment, and added to that balance.

If the annual instalment is P dollars, prove that:

- the amount owing at the beginning of the second year of the loan is $(1100 - P)$ dollars. 2
- the amount owing at the beginning of the third year of the loan is $(1210 - 2.1P)$ dollars. 2
- if the loan (including interest charges) is exactly repaid at the end of n years, then 4

$$P = \frac{100(1.1)^n}{(1.1)^n - 1}$$

(b) (i) Amount owing at the beginning of the second year.

$$A_1 = 1000 \left(1 + \frac{10}{100} \right) - P \checkmark$$

$$= 1000(1.1) - P \checkmark$$

$$= 1100 - P$$

(ii) Amount owing at the beginning of the third year.

$$A_2 = (1000(1.1) - P)1.1 - P \checkmark$$

$$= 1000(1.1)^2 - P(1.1 + 1) - P \checkmark$$

$$= 1210 - 2.1P$$

(iii) Continuing from part (ii)

Amount owing after 3 years

$$A_3 = [1000(1.1)^2 - P(1.1 + 1)](1.1) - P \checkmark$$

$$= 1000(1.1)^3 - P(1.1 + 1.1 + 1.1^2) \checkmark$$

continuing the pattern

Amount owing after n years

$$= 1000(1.1)^n - P(1.1 + 1.1^2 + \dots + 1.1^{n-1}) \checkmark$$

or can use $= 1000(1.1)^n - P(1.1 + 1.1^2 + \dots + 1.1^{n-1})$

$1.1 + 1.1^2 + \dots + 1.1^{n-1}$ is a geometric series with $a = 1.1$ $r = 1.1$

If after n years the loan is repaid

$$1000 \times (1.1)^n - P \left(\frac{(1.1)^n - 1}{1.1 - 1} \right) = 0 \checkmark$$

$$1000 \times (1.1)^n - 10P \left[\frac{(1.1)^n - 1}{1.1 - 1} \right] = 0$$

$$10P \left[\frac{(1.1)^n - 1}{1.1 - 1} \right] = 1000 \times (1.1)^n \checkmark$$

$$P = \frac{1000 \times (1.1)^n}{10 \left[\frac{(1.1)^n - 1}{1.1 - 1} \right]}$$

$$= \frac{100 \times (1.1)^n}{(1.1)^n - 1}$$



Part A: Loan repayments series 2

Q24

F
O
R
T
S
T

Samantha has retired with savings of \$400 000. She intends to live off this money by taking equal monthly instalments of \$M at the start of each month, starting immediately.

Samantha receives 6% p.a. interest on her account balance at the end of each month.

Let A_n be her account balance at the end of each month.

- Show that $A_2 = 400000(1.005)^2 - M(1.005)^2 - M(1.005)$
- Show that $A_n = 400000(1.005)^n - M[201(1.005^n - 1)]$
- Samantha would like her money to last 25 years. What monthly instalment does Samantha take?

1
3
2

(i) Solution

Note: the instalment is taken at the start of each month

$$\begin{aligned} A_1 &= (400000 - M)1.005 \\ &= 400000(1.005) - M(1.005) \end{aligned}$$

$$\begin{aligned} A_2 &= [400000(1.005) - M(1.005) - M]1.005 \\ &= 400000(1.005)^2 - M(1.005)^2 - M(1.005) \end{aligned}$$

(ii) Solution

$$\begin{aligned} A_2 &= 400000(1.005)^2 - M(1.005)^2 - M(1.005) \\ A_3 &= 400000(1.005)^3 - M(1.005)^3 - M(1.005)^2 - M(1.005) \\ A_n &= 400000(1.005)^n - M(1.005)^n - M(1.005)^{n-1} - \dots - M(1.005) \\ &= 400000(1.005)^n - M[1.005^n + 1.005^{n-1} + \dots + 1.005] \\ &= 400000(1.005)^n - M \left[\frac{1.005(1.005^n - 1)}{1.005 - 1} \right] \\ &= 400000(1.005)^n - M[201(1.005^n - 1)] \end{aligned}$$

(iii) Solution

After 25 years Samantha would have \$0 remaining $\Rightarrow A_{300} = 0$

$$400000(1.005)^{300} - M[201(1.005^{300} - 1)] = 0$$

$$M[201(1.005^{300} - 1)] = 400000(1.005)^{300}$$

$$M = \frac{400000(1.005)^{300}}{201(1.005^{300} - 1)} = \$2564.38$$

Q25

F
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Georgina borrows \$650 000 to purchase her first home. She takes out a loan over 30 years, to be repaid in equal monthly instalments. The interest rate is 5.4% per annum reducible, calculated monthly.

- Show that the amount, A_n , owing after the n th repayment is given by the formula:

1

$$A_n = 650000(1.0045)^n - M(1 + 1.0045 + 1.0045^2 + \dots + 1.0045^{n-1})$$

- Find the monthly repayment required to repay the loan in 30 years.
- Georgina wants pay the loan off in less than 30 years. If she can afford to pay \$5 000 per month, how many months will it take her to pay off the home loan?
- How much will Georgina save in interest if she pays \$5 000 per month?

2

2

1

$$(i) P = \$650000 \quad r = 5.4 \div 100 \div 12 = 0.0045$$

$$A = P(1+r)^n - M$$

$$A_1 = 650000(1.0045)^1 - M$$

$$A_2 = A_1(1.0045)^1 - M$$

$$A_2 = [650000(1.0045)^1 - M](1.0045) - M$$

$$A_2 = 650000(1.0045)^2 - M(1.0045) - M$$

$$A_2 = 650000(1.0045)^2 - M[1 + 1.0045]$$

$$A_3 = (650000(1.0045)^2 - M[1 + 1.0045])(1.0045) - M$$

$$A_3 = 650000(1.0045)^3 - M[1 + 1.0045 + 1.0045^2]$$

...

...

...

$$A_n = 650000(1.0045)^n - M[1 + 1.0045 + \dots + 1.0045^{n-1}]$$

(ii)

$$\text{Months} = 30 \times 12 = 360 \text{ repayments}$$

$$A_{360} = 0 \text{ (loan repaid)}$$

$$A_n = 650000(1.0045)^n - M[1 + 1.0045 + \dots + 1.0045^{n-1}]$$

$$0 = 650000(1.0045)^n - M[1 + 1.0045 + \dots + 1.0045^{n-1}]$$

$$M[1 + 1.0045 + \dots + 1.0045^{n-1}] = 650000(1.0045)^n$$

$$M = \frac{650000(1.0045)^n}{1 + 1.0045 + \dots + 1.0045^{n-1}}$$

$$\frac{1 + 1.0045 + \dots + 1.0045^{n-1}}{r}$$

The denominator is a geometric series with $a = 1$, $r = 1.0045$ and $n = 360$

$$S_n = \frac{a(r^n - 1)}{r}$$

$$S_{360} = \frac{1((1.0045)^{360} - 1)}{0.0045}$$

$$S_{360} = \frac{(1.0045)^{360} - 1}{0.0045}$$

$$\therefore M = \frac{(650000(1.0045)^{360}) \times 0.0045}{(1.0045)^{360} - 1}$$

$$M = \$3649.95$$

$$(iii) A_n = 650000(1.0045)^n - 5000S_n$$

$$A_n = \$0 \text{ paid off}$$

$$5000S_n = 650000(1.0045)^n$$

$$5000 \left[\frac{(1.0045)^n - 1}{0.0045} \right] = 650000(1.0045)^n$$

$$5000(1.0045)^n - 5000 = 2925(1.0045)^n$$

$$5000(1.0045)^n - 2925(1.0045)^n = 5000$$

$$(1.0045)^n[5000 - 2925] = 5000$$

$$(1.0045)^n = \frac{5000}{2075}$$

$$\ln(1.0045^n) = \ln \left[\frac{5000}{2075} \right]$$

$$n \ln(1.0045) = \ln \left[\frac{5000}{2075} \right]$$

$$n = \frac{\ln \left[\frac{5000}{2075} \right]}{\ln 1.0045}$$

$$n = 195.88$$

$$= 196 \text{ months}$$

(iv) Total of loan over 30 years

$$360 \times \$3649.95 = \$1313982$$

$$\text{Total of loan by paying } \$5000/\text{month}$$

$$196 \times \$5000 = \$980000$$

$$\text{Interest Saving}$$

$$\$1313982 - \$980000 = \$333982$$

Q26

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N

A farmer borrows \$80 000 to purchase new machinery. The interest is calculated monthly at the rate of 2% per month, and is compounded each month.

The farmer intends to repay the loan with interest in two equal annual instalments of \$ M at the end of the first and second years.

- How much does the farmer owe at the end of the first month?
- Write an expression involving M for the total amount owed by the farmer after 12 months, just after the first instalment of \$ M has been paid.
- Find an expression for the amount owed at the end of the second year and deduce that

$$M = \frac{80\,000 \times (1.02)^{24}}{(1.02)^{12} + 1}$$

- What is the total interest over the two year period?

(b) (cc) A_n is amount owed at end of n months

P is principal, amount borrowed.

in this case $P = 80\,000$

(ci) $A = P \left(1 + \frac{r}{100}\right)^n$

$$A_1 = 80\,000 \times \left(1 + \frac{2}{100}\right)^1$$

$$\text{ie. } A_1 = 80\,000 \times 1.02 = \$81\,600 \quad (1m)$$

$$\begin{aligned} \text{(cii)} \quad A_2 &= A_1 \left(1 + \frac{2}{100}\right) \\ &= A_1 \times 1.02 \\ &= (80\,000 \times 1.02) \times 1.02 \\ A_2 &= 80\,000 \times 1.02^2 \end{aligned}$$

$$\begin{aligned} A_3 &= A_2 \times 1.02 \\ &= (80\,000 \times 1.02^2) \times 1.02 \\ A_3 &= 80\,000 \times 1.02^3 \end{aligned}$$

Following this pattern for first 12 months, then:

$$A_{12} = 80\,000 \times 1.02^{12}$$

So after 1st instalment of \$ M ,
Amount owing after 12 months
 $= 80\,000 \times 1.02^{12} - M$ (2m)

(iii) 1st repayment made after A_{12} .
But 2nd repayment not made until after A_{24} .

$$\begin{aligned} A_{13} &= A_{12} \left(1 + \frac{2}{100}\right) \\ &= A_{12} \times 1.02 \end{aligned}$$

$$A_{13} = [80\,000 \times 1.02^{12} - M] \times 1.02$$

$$A_{14} = [80\,000 \times 1.02^{12} - M] \times 1.02^2$$

$$A_{15} = [80\,000 \times 1.02^{12} - M] \times 1.02^3$$

Following this pattern:

$$A_{24} = [80\,000 \times 1.02^{12} - M] \times 1.02^{12}$$

2nd (Final) repayment of \$ M is paid after A_{24} :

$$\text{ie. } [80\,000 \times 1.02^{12} - M] \times 1.02^{12} - M$$

But after 2nd/Final repayment there is no money owing.

$$\therefore [80\,000 \times 1.02^{12} - M] \times 1.02^{12} - M = 0$$

$$80\,000 \times 1.02^{24} - 1.02^{12} M - M = 0$$

$$80\,000 \times 1.02^{24} - M(1.02^{12} + 1) = 0$$

$$\text{ie. } M(1.02^{12} + 1) = 80\,000 \times 1.02^{24}$$

$$\therefore M = \frac{80\,000 \times 1.02^{24}}{(1.02^{12} + 1)}$$

$$\text{ie. } M = \frac{80\,000 \times (1.02)^{24}}{(1.02^{12} + 1)} \quad (2m)$$

$$\therefore M = \frac{80\,000 \times 1.02^{24}}{(1.02^{12} + 1)}$$

$$\text{ie. } M = \frac{80\,000 \times (1.02)^{24}}{(1.02^{12} + 1)}$$

$$\therefore M = \frac{80\,000 \times 1.02^{24}}{(1.02^{12} + 1)}$$

(iv) From calculator

$$M = \frac{80\,000 \times 1.02^{24}}{1.02^{12} + 1} = 56\,728.95203$$

$$\text{ie. each instalment } M = \$56\,728.95$$

$$\therefore \text{Total repaid in the 2 instalments}$$

$$= 2 \times \$56\,728.95$$

$$= \$113\,457.90$$

$$\therefore \text{Interest} = (\text{Total repaid}) - (\text{original loan of } \$80\,000)$$

$$= \$113\,457.90 - \$80\,000$$

$$= \$33\,457.90$$



Part A: Loan repayments series 2



Q27

An amount of \$10 000 is borrowed and an interest rate of 1% per month is charged monthly. An amount M is repaid every month.

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- (i) If A_n is the amount owing after n months, show that $A_n = \$10000(1.01)^n - M \left(\frac{1.01^n - 1}{0.01} \right)$. 4
- (ii) Find the value of M , to the nearest cent, if the loan is repaid at the end of 5 years. 2
- (iii) How much extra, in total, will be repaid if the loan is taken over 7 years? 2

(c)

(i) \$10000, $t = 1.01$

Amount owing after 1 month,
 A_1 is given by
 $A_1 = \$10000 \times 1.01 - M$

Amount owing after 2 months,
 A_2 is given by
 $A_2 = A_1(1.01) - M$
 $= [\$10000(1.01) - M] \cdot 1.01 - M$
 $= \$10000(1.01)^2 - M(1 + 1.01)$

Amount owing after 3 months,
 A_3 is given by
 $A_3 = A_2(1.01) - M$
 $= [\$10000(1.01)^2 - M(1 + 1.01)] \times 1.01 - M$
 $= \$10000(1.01)^3 - M(1 + 1.01 + 1.01^2)$

$\therefore A_n = \$10000(1.01)^n - M(1 + 1.01 + \dots + 1.01^{n-1})$

Now $1 + 1.01 + 1.01^2 + \dots + 1.01^{n-1}$
is a geometric series with
 $a = 1$, $r = 1.01$, n terms

Now $S = a \frac{(r^n - 1)}{r - 1}$

$\therefore S = 1 \frac{(1.01^n - 1)}{1.01 - 1}$

$A_n = \$10000(1.01)^n - M \frac{(1.01^n - 1)}{0.01}$

(ii) $A_{60} = 0$

$\therefore \$10000(1.01)^{60} = M \frac{(1.01^{60} - 1)}{0.01}$

$\therefore M = \frac{\$10000(1.01)^{60} \times 0.01}{1.01^{60} - 1}$
 $= \$222.44 \text{ \$ } \textcircled{2}$

(iii) $M = \frac{\$10000(1.01)^{84} \times 0.01}{1.01^{84} - 1}$
 $= \$176.53$

$\therefore$ repayments over 7 years
 $= \$176.53 \times 84$
 $= \$14828.52$

repayments over 5 years
 $= \$222.44 \times 60$
 $= \$13346.40$

$\therefore$ extra $= \$14828.52 - \13346.40
 $= \$1482.12 \text{ \$ } \textcircled{2}$

Q28

Alex borrowed \$60 000 to buy a small business. He was charged 6% p.a. compounding monthly on the balance owing and he repaid the loan plus interest in equal monthly repayment over 5 years.

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L
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M
A
R

- (i) Show that Alex owed \$ $(60\,300 - M)$ immediately after making his first monthly repayment of \$ M . 1
- (ii) Show that Alex owed \$ $[60\,000(1.005)^3 - M(1.005^2 + 1.005 + 1)]$ immediately after he made three monthly repayments. 2
- (iii) Calculate his monthly repayment, \$ M to the nearest five cents. 2
- (iv) Calculate the total amount of interest paid. 1

d) $r = 0.06 \div 12 = 0.005$

i) $A_1 = P(1+r)^1 - M$
 $= 60000(1.005)^1 - M$
 $= \$ (60\,300 - M)$

ii) $A_3 = (P + I) - (R + I)$
 $P + I = 60000(1.005)^3$
 $R + I = M + M(1.005)^1 + M(1.005)^2$
 $= M(1.005^2 + 1.005 + 1)$
 $\therefore A_3 = \$ (60000(1.005)^3 - M(1.005^2 + 1.005 + 1))$

iii) $A_{60} = 0$
 $0 = 60000(1.005)^{60} - M \frac{(1.005^{60} - 1)}{0.005}$
 $M(1.005^{60} - 1) = 60000(1.005)^{60} \times 0.005$
 $\therefore M = \$1159.95$

iv) Interest $= (1159.95 \times 60) - 60000$
 $= \$9597$

Q 1

ACE

Colin invests \$P at 8% per annum compounded annually. He plans to withdraw \$5000 at the end of each year for six years to cover university fees.

- Write down an expression for the amount A_1 remaining in the account following the withdrawal of the first \$5000.
- Find an expression for the amount A_3 remaining in the account after the third withdrawal.
- How much does Colin need to invest if the account balance is to be \$0 at the end of the six years?

1

2

2

15(c) (i)	$A = P(1+r)^n$ $= P \times (1+0.08)^1$ $= P(1.08)$ After 1 year $A_1 = P(1.08) - 5000$	1 Mark: Correct answer.
15(c) (ii)	After 2 years $A_2 = (P(1.08) - 5000) \times 1.08 - 5000$ $= P \times 1.08^2 - 5000(1.08 + 1)$ After 3 years $A_3 = [(P \times 1.08^2 - 5000(1.08 + 1)) \times 1.08 - 5000]$ $= P \times 1.08^3 - 5000(1.08^2 + 1.08 + 1)$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
15(c) (iii)	After 6 years $A_6 = P \times 1.08^6 - 5000(1.08^5 + 1.08^4 + \dots + 1)$ To find P when $A_6 = 0$ $0 = P \times 1.08^6 - 5000 \left(\frac{1.08^6 - 1}{1.08 - 1} \right)$ $0 = P \times 1.08^6 \times 0.08 - 5000(1.08^6 - 1)$ $P \times 1.08^6 \times 0.08 = 5000(1.08^6 - 1)$ $P = \frac{5000(1.08^6 - 1)}{1.08^6 \times 0.08}$ $= \$23\,114.40$ Colin needs to invest \$23 114.40	2 Marks: Correct answer. 1 Mark: Obtains a correct expression for A_6 and uses $A_6 = 0$.

Q 2

A
B
B
O
T
S
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I
G
H

Jane invests \$400 in a bank account at the beginning of each month for 8 years. Interest is to be paid at a rate of 6% per annum compounded monthly.

- Find the amount in the account at the end of the first month.
- Show that the total value of her investment at the end of n years is given by $\$400(1.005 + 1.005^2 + 1.005^3 + \dots + 1.005^{12n})$
- Find the final value of Jane's investment at the end of the 8 years. Give your answer correct to the nearest dollar.
- What single investment at the beginning of the 8 years, with interest compounded monthly, would achieve the same final value? Give your answer to the nearest dollar.

1

1

3

2

(c)(i) $A = P(1+r)^n$
 $r = 0.06 \div 12$
 $= 0.005$
 $A_1 = 400(1.005)^1$
 $= 400(1.005)$
 (ii) $A_2 = 400 \times 1.005^2 + 400 \times 1.005$
 $n \text{ years} = 12n \text{ months}$
 $\therefore A_n = 400 \times 1.005^{12n} + \dots + 400 \times 1.005^3 + 400 \times 1.005^2 + 400 \times 1.005$

(iii) $A_8 = 400 \times 1.005 + \dots + 400 \times 1.005^{96}$
 $= 400(1.005 + \dots + 1.005^{96})$
 $\uparrow$
 G.P, $a = 1.005$, $r = 1.005$,
 $n = 96$
 $S_n = \frac{a(r^n - 1)}{r - 1}$
 $\therefore A_8 = \frac{400 \times 1.005(1.005^{96} - 1)}{1.005 - 1}$
 $= \$49\,377$

(iv) $A = P(1+r)^n$
 $49377 = P(1.005)^{96}$
 $P = 49377 \div 1.005^{96}$
 $= \$30\,590$
 $\therefore \text{Single amount is } \$30\,590$

Q 3

A
B
B
O
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S
L
E
I
G
H

Mary decided to contribute to a superannuation fund by investing \$3000 each year at the beginning of 1990. The interest was paid at a rate of 7% p.a. for 10 years after which it was changed to 8% p.a. starting January 1, 2000. She plans to continue to invest in this fund until the year 2010.

- Show that after 10 years her investment is worth \$44 350.80.
- How much is her investment worth at the end of the 11th year?
- How much is her investment worth at the end of 2010?

2
1
2

(b) end 1990 1st 3000 amounts to $3000(1.07)^1$
 1991 2nd 3000 " " $3000(1.07)^2$
 1999 10th 3000 " " $3000(1.07)^{10}$
 geometric series $3000(1.07)^1 + 3000(1.07)^2 + \dots + 3000(1.07)^{10}$

$$\text{Total} = \frac{a(r^n - 1)}{r - 1} = \frac{3000(1.07)(1.07^{10} - 1)}{1.07 - 1}$$

$$= \$44\,350.80$$

(ii) At end of 11th year

$$\text{Amt} = 44\,350.80(1.08) + 3000(1.08)$$

$$= \$51\,138.86$$

(or $47\,350.8 \times 1.08$)

(iii) \$44 350.80 earns 8% compound interest for 11 years (begin - 2010)
 \$3000 invested every year at 8% for 11 years

$$\text{Total} = 44\,350.80(1.08)^{11} + \frac{3000(1.08)(1.08^{11} - 1)}{1.08 - 1}$$

$$= 103\,410.05 + 53\,931.38$$

$$= \$157\,341.43$$

Q 4

A
C
E

George is saving for a holiday. He opens a savings account with an interest rate of 0.4% per month compounded monthly at the end of each month. George decides to deposit \$450 into the account on the first of each month. He makes his first deposit on the 1st December 2011 and his last on the 1st June 2014. George withdraws the entire amount, plus interest, immediately after his final interest payment on the 30th June 2014.

- How much did George deposit into his saving account? Answer correct to the nearest dollar. 1
- How much did George withdraw from his account on the 30th June 2014? Answer correct to the nearest dollar. 3
- George's holiday is postponed due to family illness. He decides to deposit \$12 000 into a different account with an interest rate of 5% p.a. compounded quarterly for 2 years. How much will George receive at the end of the investment period? Answer correct to the nearest dollar. 2

16(a) (i)	31 deposits between 1 st December 2011 and 1 st June 2014. Total deposited = 450×31 $= \$13950$	1 Mark: Correct answer.
16(a) (ii)	1 st deposit - $A = P(1+r)^n$ $= 450(1+0.004)^{31}$ $S = 450(1.004) + 450(1.004)^2 + 450(1.004)^3 + \dots + 450(1.004)^{31}$ G.P. with $a = 450(1.004)$, $r = 1.004$ and $n = 31$ $s = \frac{450(1.004)[1.004^{31} - 1]}{1.004 - 1}$ $= \$14879.57127$ $\approx \$14880$ George withdraws \$14880 from his account.	3 Marks: Correct answer. 2 Marks: Identifies G.P. with 31 terms. 1 Mark: Uses compound interest formula with one correct value.
16(a) (iii)	$P = \$12000$, $r = \frac{0.05}{4} = 0.0125$ and $n = 2 \times 4 = 8$ $A = P(1+r)^n$ $= 12000(1+0.0125)^8$ $= \$13253.83321$ $\approx \$13254$ George will receive \$13254 after 2 years.	2 Marks: Correct answer. 1 Mark: Uses compound interest formula with one correct value.



Q 5

B
A
R
K
E
R

Jessica plans to work for the next ten years.

During that time she wants to save \$50 000.

She decides to invest a fixed amount of money at the beginning of each month during this time.

Interest is paid at a fixed rate of 6% p.a. **compounded monthly**.

(i) Let M be the monthly investment in dollars.

Show that the total investment R after 10 years is given by

$$R = M[(1.005) + (1.005)^2 + (1.005)^3 + \dots + (1.005)^{120}]$$

(ii) Find the amount M needing to be deposited each month to reach her goal.

$$50000 \quad 6\% \text{ p.a.} = 0.5\% \text{ per month}$$

$$(i) A_1 = M \times 1.005$$

$$\begin{aligned} A_2 &= (A_1 + M) \times 1.005 \\ &= (1.005M + M) \times 1.005 \\ &= 1.005^2 M + 1.005M \end{aligned}$$

$$\begin{aligned} A_3 &= (A_2 + M) \times 1.005 \\ &= ((1.005^2 M + 1.005M) + M) \times 1.005 \\ &= 1.005^3 M + 1.005^2 M + 1.005M \end{aligned}$$

$$\begin{aligned} A_{120} = R &= 1.005^{120} M + 1.005^{119} M + \dots + 1.005M \\ &= M(1.005 + 1.005^2 + \dots + 1.005^{119} + 1.005^{120}) \end{aligned}$$

$$b)(ii) \quad 1.005 + 1.005^2 + \dots + 1.005^{120}$$

$$\begin{aligned} a &= 1.005 \quad r = 1.005 \quad n = 120 \\ R &= M \times \left(\frac{1.005(1.005^{120} - 1)}{1.005 - 1} \right) \end{aligned}$$

$$50000 \times (1.005 - 1) = M(1.005(1.005^{120} - 1))$$

$$\begin{aligned} M &= \frac{50000 \times (1.005 - 1)}{1.005(1.005^{120} - 1)} \\ &= 303.5845868 = \$303.58 \end{aligned}$$

Q 6

B
A
R
K
E
R

A retirement fund pays 8% per annum compound interest on the money invested in it. What investment must a worker make at the beginning of each year if he wishes to retire with a lump sum of \$200 000 after 25 years (with his last investment at the beginning of the 25th year)?

$$\begin{aligned} d. \quad 1^{st} \quad A_1 &= M \times (1.08)^{25} \\ 2^{nd} \quad A_2 &= M \times 1.08^{25} + M \times 1.08^{24} \\ &\vdots \\ 200000 &= M(1.08 + 1.08^2 + \dots + 1.08^{25}) \\ 200000 &= M \times 1.08(1.08^{25} - 1) \\ &\quad 0.08 \\ M &= \frac{200000 \times 0.08}{1.08(1.08^{25} - 1)} \\ M &= 2533.107 \\ M &= \$2533.11 \end{aligned}$$



Part B: Superannuation series 2



Q 7

B
A
R
K
E
R

David invested \$1200 on 1st January every year.
He was paid 6.5% per annum interest compounded annually.

- How much was the investment worth at the end of 15 years?
- How many years in total would it take until the accumulated value of the investment was \$64 000?

$$A = 1200 [1.065 + 1.065^2 + \dots + 1.065^{15}]$$

$$= 1200 \left[\frac{1.065(1.065^{15} - 1)}{0.065} \right]$$

$$= \$30\,904.81$$

c) ii)

$$1200 \left[\frac{1.065(1.065^n - 1)}{0.065} \right] = 64\,000$$

$$1.065^n - 1 = \frac{64\,000 \times 0.065}{1200 \times 1.065}$$

$$1.065^n = 4.255086072$$

$$n = \log_{1.065} (4.255086072)$$

$$n = 22.995$$

$$\therefore n = 23 \text{ years}$$

Q 8

B
A
U
L
K
H
A
M

Karen contributes to a superannuation fund. She contributes \$250 at the start of every quarter. The investment pays 8% pa interest, compounding quarterly. She continues making contributions for 30 years.

- How much does she contribute altogether?
- What is the value of her initial \$250 investment at the end of 30 years?
- Find the total value of her superannuation.

@ 2% per quarter ; terms 30 years = 120 quarters.

$$(i) 250 \times 120 = \$30,000 \quad \checkmark$$

$$(ii) A_n = P \left(1 + \frac{r}{100}\right)^n$$

$$= 250 \left(1 + \frac{2}{100}\right)^{120}$$

$$= \$2691.29 \quad \checkmark$$

$$\text{1st instalment is worth} = 250(1.02)^{120}$$

$$\text{2nd} = 250(1.02)^{119}$$

$$\text{last (120th instalment)} = 250 \times (1.02)^1$$

$$\text{Superannuation} = 250 \times 1.02 + 250 \times 1.02^2 + \dots + 250 \times 1.02^{120}$$

This is a GP for which $S_n = a \frac{(r^n - 1)}{r - 1}$

$$\therefore S_{120} = 250 \times 1.02 \times \frac{(1.02^{120} - 1)}{1.02 - 1} \quad \checkmark$$

$$= \$124\,505.83 \quad \checkmark$$

Q 9

B
A
U
L
K
H
A
M

When Susan was born, her father deposited \$200 into a Trust account earning 12% p.a. interest compounding annually. The interest is paid on her birthday each year. Susan's father decided to deposit an additional \$200 into this account on her birthday each year, immediately after the interest is received.

- (i) Find the value that the initial deposit would amount to on her 18th birthday. 1
(ii) Let A_n = total amount in her account on her n th birthday 3
Show that $A_n = \frac{5000(1.12^{n+1}-1)}{3}$
(iii) Susan's father made his last deposit of \$200 on her 17th birthday. On her eighteenth birthday, he gave all the money in the account to Susan. How much money did she receive? 2

$$\begin{aligned} \text{(i)} \quad A &= 200(1.12)^{18} = \$1537.99 \quad (1) \\ \text{(ii)} \quad A_1 &= 200(1.12) + 200 \\ A_2 &= A_1 \times 1.12 + 200 \\ &= 200(1.12)^2 + 200(1.12) + 200 \quad \text{Develop} \\ A_3 &= A_2 \times 1.12 + 200 \quad (1) \\ &= 200(1.12)^3 + 200(1.12)^2 + 200(1.12) + 200 \\ &= 200(1.12^3 + 1.12^2 + 1.12 + 1) \end{aligned}$$

Continuing the pattern:

$$\begin{aligned} A_n &= 200(1.12^n + 1.12^{n-1} + \dots + 1.12 + 1) \quad (1) \\ &= 200 \times \frac{1(1.12^{n+1} - 1)}{1.12 - 1} \quad \text{geom series} \\ &= 200 \times \frac{(1.12^{n+1} - 1)}{0.12} \quad a=1, r=1.12 \\ &= \frac{5000(1.12^{n+1} - 1)}{3} \quad (1) \quad \text{(n+1) term} \quad S_n = \frac{a(r^{n+1}-1)}{r-1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \text{Final amt} &= A_{17} \times 1.12 \quad \text{or} \quad A_{18} - 200 \\ &= \$11150 \times 1.12 \quad \text{or} \quad \$12688 - 200 \\ &= \$12488 \quad (2) \\ &\quad -1 \text{ if out by } 200 \end{aligned}$$

Q10

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K

Gayle has a superannuation fund, which pays 5% per annum interest compounding annually. Gayle pays \$12 000 into the fund on 1 July each year.

- (i) What is the value of Gayle's superannuation fund on 30 June one year after she makes her first payment? 1
(ii) What is the value of Gayle's superannuation fund on 30 June ten years after she makes her first payment? 3
(iii) After making her tenth payment, Gayle considers increasing her payment to M dollars per year. Show that if Gayle does this, then the value of her superannuation fund twenty years after her first payment of \$12 000 was made, would be approximately given by 3

$$13.2068(12\,000 \times 1.05^{10} + M).$$

$$\text{b) } A_1 = 12000(1.05)$$

$$\text{(i)} \quad \$12\,600 \quad \checkmark$$

$$\text{(ii)} \quad A_1 + A_2 + \dots + A_{10}$$

$$= 12000(1.05 + 1.05^2 + \dots + 1.05^{10})$$

$$a = 1.05$$

$$r = 1.05$$

$$n = 10$$

$$S_n = \frac{1.05(1.05^{10} - 1)}{0.05}$$

$$= 12000 \times S_n$$

$$= \$158\,481.45 \quad \checkmark$$

$$\text{(iii)} \quad A_1 + A_2 + \dots + A_{19} + A_{20}$$

$$= 12000(1.05^{20} + 1.05^{19} + \dots + 1.05^1)$$

$$+ M(1.05^{10} + 1.05^9 + \dots + 1.05^1) \quad \checkmark$$

$$= 12000(1.05)^{10}(1.05^{10} + 1.05^9 + \dots + 1.05^1)$$

$$+ M(1.05^{10} + 1.05^9 + \dots + 1.05^1) \quad \checkmark$$

$$= (12000(1.05)^{10} + M)(1.05^{10} + \dots + 1.05^1)$$

$$n = 10$$

$$a = 1.05$$

$$r = 1.05$$

$$S_{10} = \frac{1.05(1.05^{10} - 1)}{0.05}$$

$$= (12000(1.05)^{10} + M)(13.2068) \quad \checkmark$$



Q11

Antonia and Melina recently received an inheritance windfall of \$20 000 each. After speaking to a financial adviser, both decide to invest their money with 'Gauss Investments Australia'.

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Antonia decides to invest the \$20 000 in a savings policy at 7.5% p.a. and at the end of the first month she makes a contribution of \$180 to this policy. Antonia continues to contribute \$180 per month at the same interest rate. The savings policy is compounded monthly with interest paid just before each contribution.

Let S_n be the amount Antonia's savings policy grows to by the end of the n th contribution.

- (i) Show that after the second contribution, 2
 $A_2 = 20\,000(1.00625)^2 + 180(1 + 1.00625)$.
- (ii) Antonia's intention is to collect her savings when the policy reaches \$50 000. For how many months must Antonia contribute to reach her goal? 2

Melina decides to invest her \$20 000 in a term deposit for fifteen years with interest compounded monthly.

- (iii) At what annual interest rate must Melina invest her money if she is to also receive \$50 000? 2

(b)(i) (2 marks)

$$\begin{aligned} A_1 &= 20\,000(1.00625) + 180 \\ A_2 &= A_1(1.00625) + 180 \\ &= (20\,000(1.00625) + 180)(1.00625) + 180 \\ &= 20\,000(1.00625)^2 + 180(1 + 1.00625) \end{aligned}$$

(b)(ii) (2 marks)

Let $A_n = 50\,000$.

$$\therefore 50\,000 = 20\,000(1.00625)^n + 180(1 + 1.00625 + \dots + 1.00625^{n-1})$$

$$50\,000 = 20\,000(1.00625)^n + 180 \left[\frac{(1.00625)^n - 1}{0.00625} \right]$$

$$50\,000 = 20\,000(1.00625)^n + 28\,800(1.00625)^n - 28\,800$$

$$78\,800 = 48\,800(1.00625)^n$$

$$(1.00625)^n = \frac{788}{488}$$

$$\log_e(1.00625)^n = \log_e\left(\frac{788}{488}\right)$$

$$n = 76.9 \text{ months}$$

$$= 77 \text{ months}$$

$\therefore$ after 77 months/payments Antonia will have reached her goal.

(b)(iii) (2 marks)

Using the formula: $A = P(1+r)^n$ where $n = 15 \times 12$

$$50\,000 = 20\,000(1+r)^{180}$$

$$\frac{5}{2} = (1+r)^{180}$$

$$(1+r) = \left(\frac{5}{2}\right)^{\frac{1}{180}}$$

$$1+r = 1.005103$$

$$r = 0.005103$$

$$= 0.51\% \text{ per month}$$

$$= 6.12\% \text{ p.a.}$$

Q12

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Kenny begins his retirement with \$500 000 at the beginning of 2009. The annual interest rate is 8% p.a. Interest is calculated annually on the balance at the beginning of the year and added to the remaining balance. Kenny plans to withdraw \$56 000 annually, with the first withdrawal at the end of 2009.

Let A_n be the remaining balance after the n th withdrawal.

- (i) Show that $A_2 = (5 \times 10^5)(1.08)^2 - 5.6 \times 10^4(1.08 + 1)$. 2
- (ii) Show that $A_n = 10^5[7 - 2(1.08)^n]$. 2
- (iii) In which year will Kenny's fund reach zero? 3

(a) (i) (2 marks)

$$A_1 = 500\,000(1.08) - 56\,000$$

$$= 5 \times 10^5(1.08) - 5.6 \times 10^4$$

$$A_2 = A_1(1.08) - 5.6 \times 10^4$$

$$= [5 \times 10^5(1.08) - 5.6 \times 10^4](1.08) - 5.6 \times 10^4$$

$$= 5 \times 10^5(1.08)^2 - 5.6 \times 10^4(1.08 + 1) \text{ as required.}$$

(a)(ii) (2 marks)

$$A_n = 5 \times 10^5(1.08)^n - 5.6 \times 10^4(1 + 1.08 + \dots + 1.08^{n-1})$$

$$= 5 \times 10^5(1.08)^n - 5.6 \times 10^4 \left[\frac{1.08^n - 1}{0.08} \right]$$

$$= 5 \times 10^5(1.08)^n - 7 \times 10^5(1.08^n - 1)$$

$$= 5 \times 10^5(1.08)^n - 7 \times 10^5(1.08)^n + 7 \times 10^5$$

$$= 7 \times 10^5 - 2 \times 10^5(1.08)^n$$

$$= 10^5[7 - 2(1.08)^n] \text{ as required.}$$

(a) (iii) (3 marks)

When $A_n = 0$, the fund will have reached zero.

$$0 = 10^5[7 - 2(1.08)^n]$$

$$0 = [7 - 2(1.08)^n]$$

$$2(1.08)^n = 7$$

$$n = \frac{\log_e 3.5}{\log_e(1.08)}$$

$$= 16.27788\dots$$

During the 17th year the fund will reach zero.

Therefore during 2025 the fund will reach zero.



Q13

Monica invests \$250 into an account at the Bank of Newton. She invests the money at the beginning of each month for n years. Interest is to be paid at a rate of 6%p.a. compounded monthly.

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- Show that the total value of her investment (A_n) at the end of n years is given by $A_n = \$250(1.005 + 1.005^2 + \dots + 1.005^{12n})$.
- Find the value of the investment at the end of 7 years.
- What single investment at the beginning of the 7 years would yield the same final value for Monica? You may assume interest is compounded monthly.

(a)(i) (2marks)

$$r = \frac{6}{12} = 0.5\% \text{ and } n \text{ years} = 12n \text{ months}$$

$$A_1 = 250 \left(1 + \frac{0.5}{100} \right) = 250(1.005)$$

$$A_2 = 250(1.005)^2, \dots, A_{12n} = 250(1.005)^{12n}$$

$$\therefore A = 250(1.005 + 1.005^2 + \dots + 1.005^{12n})$$

(a)(ii) (2marks)

$$\text{When } n = 7 \text{ years} = 7 \times 12 = 84 \text{ months}$$

$$\therefore A = 250 \times \frac{1.005(1.005^{84} - 1)}{1.005 - 1} = \$26\,148.57$$

(a)(iii) (2marks)

Sample answer:

$$26\,148.57 = P(1.005)^{84}$$

$$\therefore P = \$17\,198.83$$

Q14

A person invests \$800 at the beginning of each year in a superannuation fund. Compound interest is paid at 10% per annum on the investment. The first \$800 is to be invested at the beginning of 2016 and the last is to be invested at the beginning of 2045. Calculate to the nearest dollar:

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- The amount to which the 2016 investment will have grown by the beginning of 2046 2
- The amount to which the total investment will have grown by the beginning of 2046 2

i.

$$\begin{aligned} \$800(1.1)^{30} &= \$13,959.52\dots \text{ 1} \\ &= \$13,960 \text{ 1} \end{aligned}$$

ii.

$$\begin{aligned} &800(1.1)^{30} + 800(1.1)^{29} + \dots + 800(1.1)^1 \\ &= 800((1.1)^{30} + (1.1)^{29} + \dots + (1.1)) \end{aligned}$$

GP with $a = 1.1$, $n = 30$, $r = 1.1$

$$S_{30} = \frac{1.1(1.1^{30} - 1)}{1.1 - 1}$$

$$= \frac{1.1(1.1^{30} - 1)}{0.1}$$

$$= 11(1.1^{30} - 1) \text{ 1}$$

$$\therefore 800(11(1.1^{30} - 1)) = 144,754.7399\dots \approx \$144,755 \text{ 1}$$

Q15

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Maxamillian's daughter was born on the 1st January. On that day he opened a trust account by depositing \$250. Each year, on her birthday, he deposited \$250 into this trust fund.

He continued to do this up to and including her 17th birthday. When she turned eighteen, he collected the total amount including interest from this account and presented to her.

This account paid an interest of 6% p.a. compounded every six months.

- (i) Show the initial deposit amounted to approximately \$724.57 after 18 years. 1
(ii) How much did Maximillian give his daughter on her eighteenth birthday? 3

(c) $P = \$250$ $R = 0.06 / \text{p.a.}$
 $n = 18 \text{ years}$ $R = 0.03 / 6 \text{ months}$
 $= 36 \text{ yrs}$

1) $A_1 = P(1+R)^n$
 $= 250(1.03)^{36}$

$A_1 = \$724.57$ ✓

∴ initial deposit amounts to \$724.57.

ii) A_1 amounts to: $25(1.03)^{36}$
 A_2 amounts to: $25(1.03)^{34}$
 A_3 amounts to: $25(1.03)^{32}$

A_{16} amounts to: $250(1.03)^6$
 A_{17} amounts to: $250(1.03)^4$
 A_{18} amounts to: $25(1.03)^2$

Total amount = $250(1.03^2 + 1.03^4 + \dots + 1.03^{36})$

$S.S. a = 1.03^2$ $r = 1.03^2$ $n = 18$

$S_n = a(n-1)$ ✓

$= 1.03^2(1.03^{36} - 1)$

Total = $250 \times S_{18}$

$= 250 \times 33.06869 \dots$

$= 8267.17355 \dots$ ✓

Total = \$8267.17 (nearest \$)

Q16

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A woman contributes \$1500 each year into a superannuation fund for the first 25 years 6 of her working life. For the next fifteen years (until retirement), she decides to increase this, and invest a total of \$5000 each year. Each contribution is paid at the beginning of the year.

If the investment earns 7% p.a. paid yearly over the whole year period, how much will her investment be upon retiring? (Answer to the nearest dollar)

Co) Treat the investment in 2 parts
Part 1 \$1500 invested each year for 40 years

1st $\$1500 = \1500×1.07 39

2nd $\$1500 = \1500×1.07

↓
last $\$1500 = \1500×1.07

Part 2: For the last 15 years, the extra \$3500 (to make up the \$5000 to be invested each year for 15 years)

1st $\$3500 = \3500×1.07 14

2nd $\$3500 = \3500×1.07

↓
last $\$3500 = \3500×1.07 ✓

Total = $1500(1.07 + 1.07^2 + \dots + 1.07^{40})$ plus
 $3500(1.07 + 1.07^2 + \dots + 1.07^{15})$ ✓
 $= 1500 \times 1.07 \frac{(1.07^{40} - 1)}{1.07 - 1} + 3500 \times 1.07 \frac{(1.07^{15} - 1)}{1.07 - 1}$

$= \$32014.35 + \94108.19 ✓

$= \$414523$ (nearest dollar) ✓

(3 marks for correctly calculating one part only)

Q17

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On being retrenched from his job, Kevin receives a cash payment of \$20 000.

One year later, he receives his first annual payout of \$10 000. He continues to receive annual payouts of \$10 000 every year thereafter.

He places all of this money in his suitcase as he receives it, and spends none.

At the end of every year, just before the next payout, Kevin spends 20% of the money in his suitcase on a holiday.

Let A_n be the amount Kevin has in his suitcase immediately after his n^{th} annual payout.

- (i) Show that Kevin has \$26 000 in his suitcase immediately after his first annual payout. 1
- (ii) Show that the money in Kevin's suitcase immediately after his 3rd annual payout is given by 2
 $A_3 = 20000(0.8)^3 + 10000(1 + 0.8 + 0.8^2)$.
- (iii) Show that $A_n = 50000 - 30000(0.8)^n$. 3
- (iv) After how many years will the amount in Kevin's suitcase first exceed \$48 000? 2
- (v) What is the most money Kevin will ever have in his suitcase? 1

Q15 a) (i) $A_1 = 20000 \times 0.8 + 10000 = \$26,000$ [1]

(ii) $A_2 = A_1 \times 0.8 + 10,000$
 $= [20000 \times 0.8 + 10000] \times 0.8 + 10,000$
 $= 20000(0.8)^2 + 10000[1 + 0.8]$ ✓ Well done

$A_3 = A_2 \times 0.8 + 10000$
 $= [20000(0.8)^2 + 10000[1 + 0.8]] \times 0.8 + 10000$
 $= 20000(0.8)^3 + 10000[0.8 + 0.8^2] + 10000$ [2]
 $= 20,000(0.8)^3 + 10000[1 + 0.8 + 0.8^2]$ as required

(iii) $A_n = 20,000(0.8)^n + 10000[1 + 0.8 + 0.8^2 + \dots + 0.8^{n-1}]$ ✓
 $= 20000(0.8)^n + 10000 \left[S_n = \frac{1(1 - (0.8)^n)}{1 - 0.8} \right] \rightarrow S_n = \frac{a(1 - r^n)}{1 - r}$
 $= 20000(0.8)^n + 50000(1 - (0.8)^n)$ ✓ Mostly well done
 $= 50000 - 30000(0.8)^n$ [3]

(iv) $50000 - 30000(0.8)^n > 48000$
 $-30000(0.8)^n > -2000$
 $\therefore (0.8)^n < \frac{2}{30}$ take logs $n \log 0.8 < \log \frac{1}{15}$
 and as $\log(0.8) < 0 \therefore n > \frac{\log(\frac{1}{15})}{\log(0.8)} > 12.1359$
 Some inequality signs wrong. $\therefore n = 13 \text{ yrs}$ (required to exceed \$48,000) ✓ [2]

(v) As $n \rightarrow \infty$ $A_n = 50000 - 30000(0.8)^n$
 $0.8^n \rightarrow 0 \therefore A_n \rightarrow 50,000$ (most Kevin will ever have in suitcase) [1]

Q18

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Beth and Cathy are best friends who work in the same office. Each year on January 1, they each receive a cash bonus of \$5000. They received their first bonuses in 1997. Every year Beth invests her \$5000 in superannuation at 9% p.a. compounding interest. Each year Cathy spends her bonus on an overseas trip.

(i) Show that the expression $5000(1.09^{10} + 1.09^9 + 1.09^8 + \dots + 1.09)$ represents the amount in Beth's superannuation account on January 1, 2007, immediately before her 2007 bonus was added to the account.

(ii) Show that Beth had almost \$88 000 in her superannuation account on January 1, 2007, after her 2007 bonus was credited to her account.

Cathy decides that on January 1, 2007, she will start saving for her retirement, which will occur in 20 years' time. She would like to have the same amount that Beth will have in 20 years' time from saving her annual \$5000 bonus. Cathy's account also pays 9% p.a. compounding interest.

(iii) How much will Cathy need to save each year to have the same total amount as Beth will have in 20 years' time (i.e. including the amounts Beth invested in the first 10 years)?

(iv) How much more will Cathy have to invest over the 20 years than Beth will have invested over the 30 years?

(i) Let a_n = amount in the account at the end of n years immediately before the next addition.

$$a_1 = 5000(1.09)$$

$$a_2 = [5000(1.09) + 5000](1.09)$$

$$= 5000(1.09)^2 + 5000(1.09)$$

$$= 5000[(1.09)^2 + 1.09]$$

$$a_3 = [5000\{(1.09)^2 + 1.09\} + 5000](1.09)$$

$$= 5000[1.09^3 + 1.09^2 + 1.09]$$

$$a_n = 5000[1.09^n + 1.09^{n-1} + \dots + 1.09]$$

$$\therefore a_{10} = 5000[1.09^{10} + 1.09^9 + \dots + 1.09]$$

(ii) $A_{10} + 5000$

$$= 5000 + 5000 \times \frac{1.09(1.09^{10} - 1)}{1.09 - 1}$$

$$= \$87\,801.46$$

Beth has almost \$88 000 in her account.

(iii) In 20 more years, Beth will have

$$5000 \times \frac{1.09(1.09^{30} - 1)}{1.09 - 1} = \$742\,876.09$$

Cathy

$$742\,876.09 = A \times \frac{1.09(1.09^{20} - 1)}{1.09 - 1}$$

$$= A \times 55.7645$$

$$\therefore A = 13\,321.66$$

Cathy will need to invest \$13 321.66 each year.

(iv) $\$13\,321.66 \times 20 - \$5000 \times 30 = \$116\,433.19$

Cathy will have to invest \$116 433.19 more than Beth.

Q19

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On the 1st January each year Simone invests \$M annually into a superannuation account. The account gives interest at a rate of 5% per annum, compounded annually.

(i) Show that the value of her investment at the end of 2 years was $A_2 = 2.1525M$ dollars.

(ii) Show that the value of her investment at the end of n years was $A_n = 21(1.05^n - 1)M$ dollars.

(iii) Simone wants to retire after 30 years with a million dollars in her superannuation account. Find the amount that she must invest into her account on the 1st January each year to reach her goal. Answer to the nearest cent.

$$\begin{aligned} (d) \quad A_0 &= M \\ (1) \quad A_1 &= M \times 1.05 \\ A_2 &= (M \times 1.05 + M) \times 1.05 \\ &\quad \uparrow \quad \uparrow \\ &\quad \text{NEXT INVESTMENT} \quad \text{INTEREST} \\ A_2 &= M \times 1.05^2 + M \times 1.05 \\ A_2 &= (1.05^2 + 1.05)M \\ A_2 &= 2.1525M \end{aligned}$$

$$\begin{aligned} (ii) \quad A_3 &= (A_2 + M) \times 1.05 \\ &\quad \uparrow \quad \uparrow \\ &\quad \text{NEXT INVESTMENT} \quad \text{INTEREST} \\ &= (M \times 1.05^2 + M \times 1.05 + M) \times 1.05 \\ &= M \times 1.05^3 + M \times 1.05^2 + M \times 1.05 \\ &= (1.05^3 + 1.05^2 + 1.05)M \end{aligned}$$

$$\begin{aligned} A_n &= (1.05^n + 1.05^{n-1} + \dots + 1.05)M \\ &\quad \uparrow \\ &\quad \text{GP } n \text{ terms} \\ &\quad a = 1.05 \\ &\quad r = 1.05 \\ A_n &= \frac{a(r^n - 1)}{r - 1} M \\ &= \frac{1.05(1.05^n - 1)}{1.05 - 1} M \\ &= \frac{1.05}{0.05} (1.05^n - 1) M \\ &= 21(1.05^n - 1)M \end{aligned}$$

(iii) $A_n = 1\,000\,000 \quad n = 30$

$$1\,000\,000 = 21(1.05^{30} - 1)M$$

$$M = \frac{1\,000\,000}{21(1.05^{30} - 1)}$$

$$M = 14\,334.70$$

So each investment needs to be \$14 334.70

Q20

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D

On the 1st January 2000, Toby deposited \$15 000 into a bank account that paid interest at a fixed rate of 4% per annum compounded annually. He later decided to add \$5000 to his account on 1st January each year, starting on 1st January 2007.

- (i) Write an expression for the amount in the account on 1st January 2007 after the payment of interest and the first \$5000 deposit. 1
- (ii) How much was in Toby's account on 1st January 2016 after the payment of interest and the \$5000 deposit? 4

c) i). $A = 15000 \times 1.04^7 + 5000$

ii) $A_2 = (15000 \times 1.04^7 + 5000) \times 1.04 + 5000$
 $= 15000 \times 1.04^8 + 5000 \times 1.04 + 5000$

$A_3 = (15000 \times 1.04^8 + 5000 \times 1.04 + 5000) \times 1.04 + 5000$
 $= 15000 \times 1.04^9 + 5000 \times 1.04^2 + 5000 \times 1.04 + 5000$

⋮

$A_{10} = 15000 \times 1.04^{16} + 5000 \times 1.04^9 + 5000 \times 1.04^8 + \dots + 5000$
 $= 15000 \times 1.04^{16} + 5000(1 + 1.04 + 1.04^2 + \dots + 1.04^9)$
 $= 15000 \times 1.04^{16} + 5000 \left[\frac{1(1.04^{10} - 1)}{1.04 - 1} \right]$
 $= 28094.71869 + 5000[12.00610712]$
 $= 88125.2543$
 $= \$88125.25$

Q21

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On the birth of Fernando, his parents decide to set up an investment fund for him. They agree to deposit \$500 on the 18th September 2008 (the date of birth) and \$500 on every subsequent birthday up to and including his 21st birthday. Interest is being paid at the rate 6% p.a. to be compounded:

- i) How much will the first \$500 deposit mature to after 21 years? 2
- ii) How much will Fernando receive on his 21st birthday? 2

i) $A = 500(1.06)^{21}$
 $= \$1699.78$

ii) $\frac{500 \times 1.06(1.06^{21} - 1)}{1.06 - 1}$
 $= \$21,196.15 + \500
 $= \$26,696.15$

Q 1

A
C
E

Kristy and Josh worked out that they would save \$100 000 in five years by depositing all their combined monthly salary of \$ x at the beginning of each month into a savings account and withdrawing \$3200 at the end of each month for living expenses. The savings account paid interest at the rate of 3% p.a. compounding monthly.

- Show that at the end of the second month they have
 $\$[(1.0025^2 + 1.0025)x - 3200(1.0025 + 1)]$ in their savings account. 2
- Write down an expression for the balance in their account at the end of the five years? 1
- What is their combine monthly salary? 2

15(b) (i)	$3\% \text{ p.a.} = \frac{0.03}{12} = 0.0025 \text{ per month}$ After 1 month $A_1 = \$[x \times 1.0025 - 3200]$ After 2 months $A_2 = \$[(x \times 1.0025 - 3200 + x) \times 1.0025 - 3200]$ $= \$[(1.0025^2 + 1.0025)x - 3200(1.0025 + 1)]$	2 Marks: Correct answer. 1 Mark: Sets up the first month.
15(b) (ii)	After 5 years or 60 months $A_{60} = \$[(1.0025^{60} + \dots + 1.0025)x - 3200(1.0025^{59} + \dots + 1)]$	1 Mark: Correct answer.
15(b) (iii)	After 60 months the amount required is \$100 000 $1000000 = \$[(1.0025^{60} + \dots + 1.0025)x - 3200(1.0025^{59} + \dots + 1)]$ Use the formula for the sum of a GP. $1000000 = \$\left[1.0025 \times \frac{1.0025^{60} - 1}{1.0025 - 1}\right]x - 3200 \times \frac{1.0025^{60} - 1}{1.0025 - 1}$ $x = \left(1000000 + 3200 \times \frac{1.0025^{60} - 1}{1.0025 - 1}\right) \div \left(1.0025 \times \frac{1.0025^{60} - 1}{1.0025 - 1}\right)$ $= \$4735.03$	2 Marks: Correct answer. 1 Mark: Makes significant progress towards the solution.

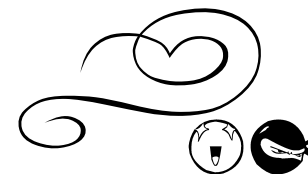
Q 2

A
C
E

Monique has set up her superannuation fund and after 10 years she has accumulated \$134 000. However due to an accident she is no longer able to work and make further contributions to the fund. Monique is leaving the money in the superannuation fund to accumulate interest at 8% p.a. compounded annually. However she needs to withdraw \$24 000 at the end of each year for normal living expenses.

- Show that at the end of the first year she has
 $\$(134000 \times 1.08 - 24000)$ in the superannuation fund. 1
- Find a similar expression for the amount in the fund after 3 years. 2
- Hence find how many years the fund will last before there is no money in it. 2

8(b) (i)	$A = P(1+r)^n$ $= 134000 \times (1+0.08)^1$ $= 134000(1.08)$ After 1 year $A_1 = 134000 \times 1.08 - 24000$	1 Mark: Correct answer.
8(b) (ii)	After 2 years $A_2 = (134000(1.08) - 24000) \times 1.08 - 24000$ $= 134000 \times 1.08^2 - 24000(1.08 + 1)$ After 3 years $A_3 = [(134000 \times 1.08^2 - 24000(1.08 + 1)) \times 1.08 - 24000]$ $= 134000 \times 1.08^3 - 24000(1.08^2 + 1.08 + 1)$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution
8(b) (iii)	After n years $A_n = 134000 \times 1.08^n - 24000(1.08^{n-1} + 1.08^{n-2} + \dots + 1)$ To find n when $A_n = 0$ $0 = 134000 \times 1.08^n - 24000 \left(\frac{1.08^n - 1}{1.08 - 1} \right)$ $134000 \times 1.08^n \times 0.08 - 24000(1.08^n - 1) = 0$ $10720 \times 1.08^n - 24000 \times 1.08^n + 24000 = 0$ $13280 \times 1.08^n = 24000$ $1.08^n = \frac{24000}{13280}$ $n \times \log 1.08 = \log \left(\frac{24000}{13280} \right)$ $n \approx 7.69 \text{ years}$	2 Marks: Correct answer. 1 Mark: Obtains a correct expression for A_n and uses $A_n = 0$



Q 3

BAULKHAM

At the beginning of the year 2008 a company bought a new machine for \$15000. Each subsequent year the value of the machine decreases by 20% of its value at the start of the year. When the value of the machine falls below \$500, the company will replace it.

To plan for a replacement machine, the company pays \$1000 at the start of each year into a savings account. The account pays a fixed rate of 5% compound interest p.a. The first payment was made when the machine was bought and the last payment will be made at the start of the year in which the machine is replaced.

- (i) What will be the value of the machine at the start of 2010? 1
- (ii) In which year will the company replace the machine? 3
- (iii) How much will the savings account be worth at the end of the year in which the machine will be replaced? 3

$$\begin{aligned} a) A_n &= 15000(0.8)^n \\ A_2 &= 15000(0.8)^2 \\ &= 9600 \end{aligned}$$

$\therefore$ machine will be worth ①
\$9600 at start of 2010

$$\begin{aligned} (ii) A_n &< 500 \\ 15000(0.8)^n &< 500 \\ (0.8)^n &< \frac{1}{30} \\ \log(0.8)^n &< \log \frac{1}{30} \\ n \log 0.8 &< \log \frac{1}{30} \end{aligned}$$

$$\begin{aligned} n &> \frac{\log \frac{1}{30}}{\log 0.8} \\ n &> 15.24219437 \end{aligned}$$

$\therefore$ company will replace the machine during 2023 ②

$$\begin{aligned} (iii) \text{ 1st investment} &= 1000(1.05)^{16} \\ \text{2nd investment} &= 1000(1.05)^{15} \\ &\vdots \end{aligned}$$

$$\text{last investment} = 1000(1.05)^1$$

$$\begin{aligned} \text{Total} &= 1000 \{ 1.05 + 1.05^2 + \dots + 1.05^{16} \} \\ &= 1000 \left[\frac{1.05(1.05^{16} - 1)}{0.05} \right] \\ &= 24840.36636 \end{aligned}$$

$\therefore$ Account will be worth \$24840.37 ③